Finite- and Infinite-Dimensional Vectors

Extended non-spiral edition

for mathematics and physics

Part 0	Geometric vectors in $\mathbb{R}^3$: $\mathbf{i}, \mathbf{j}, \mathbf{k}$
Part I	Finite-dimensional vectors over $\mathbb{R}$: $\mathbb{R}^N$
Part II	Infinite-dimensional vectors (natural index): ℓ^2
Part III	Infinite-dimensional vectors (real index): L^2
Part IV	Fourier coordinates and evolution equations

Linear algebra for mathematics and physics, with applications to partial differential equations and quantum mechanics. Short proofs and worked examples are included. Parts 0–III use the real field. Part IV introduces complex Fourier notation where it is naturally required.

How to use these notes

Part 0 recalls geometric vectors in $\mathbb{R}^3$. It also establishes the norm, Cauchy–Schwarz and projection results used later.

The main ideas are then developed in three parallel settings:

$$\boxed{\mathbb{R}^N \longrightarrow \ell^2 \longrightarrow L^2}.$$

Part I uses finitely many components v_i. Part II uses a_n, indexed by $n \in \mathbb{N}$. Part III begins with f_x, indexed by $x \in \mathbb{R}$, before returning to $f(x)$.

Matching section numbers treat matching ideas. For example, Sections 1.10, 2.10 and 3.10 treat projections; Sections 1.11, 2.11 and 3.11 compare p-norms. The final table collects the full dictionary.

Infinite-dimensional results are preparatory. Convergence, domains and spectral subtleties are stated without measure-theoretic detail.

Part IV applies the same coordinate picture to Fourier series, Fourier transforms, the heat equation and the infinite square well.

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Part 0: Geometric Vectors in $\mathbb{R}^3$

0. Geometric Vectors in $\mathbb{R}^3$

Part 0. A geometric warm-up using the familiar unit vectors **i**, **j** and **k**.

0.1. Coordinates and unit vectors

Definition 0.1.1 (Coordinate vector)**.**
Every vector in $\mathbb{R}^3$ has the form

$$\boxed{\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}}.$$

Its coordinate triple is (v_1, v_2, v_3).

The unit vectors are

$$\mathbf{i} = (1, 0, 0), \qquad \mathbf{j} = (0, 1, 0), \qquad \mathbf{k} = (0, 0, 1).$$

Worked example 0.1.2 (Displacement)**.**
The displacement

$$\mathbf{d} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$$

has coordinate triple $(3, -2, 1)$.

0.2. Length and norms

Definition 0.2.1 (Euclidean norm)**.**
The length of $\mathbf{v} = (v_1, v_2, v_3)$ is

$$\boxed{\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + v_3^2}}.$$

Definition 0.2.2 (Norm)**.**
A norm assigns a length $\|v\|$ to every vector v.

For all vectors u, v and scalars α:

(i) $\|v\| \geq 0$, with equality only for $v = 0$;

(ii) $\|\alpha v\| = |\alpha| \, \|v\|$;

(iii) $\|u + v\| \leq \|u\| + \|v\|$.

Worked example 0.2.3 (Length and normalisation)**.**

Let $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Then

$$\|\mathbf{v}\|_2 = 3, \qquad \widehat{\mathbf{v}} = \frac{1}{3}(2, -1, 2).$$

The vector $\widehat{\mathbf{v}}$ has length one.

0.3. Dot product and Cauchy–Schwarz

Definition 0.3.1 (Dot product)**.**
For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$,

$$\boxed{\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3}.$$

Moreover,

$$\|\mathbf{v}\|_2^2 = \mathbf{v} \cdot \mathbf{v}.$$

Theorem 0.3.2 (Cauchy–Schwarz inequality)**.**
For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$,

$$\boxed{|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|_2 \, \|\mathbf{v}\|_2}.$$

Proof. The result is immediate if $\mathbf{v} = 0$.

For $\mathbf{v} \neq 0$, a squared length cannot be negative:

$$0 \leq \left\| \mathbf{u} - \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|_2^2} \mathbf{v} \right\|_2^2.$$

Expanding gives

$$0 \leq \|\mathbf{u}\|_2^2 - \frac{(\mathbf{u} \cdot \mathbf{v})^2}{\|\mathbf{v}\|_2^2}.$$

Rearranging proves the inequality. □

Corollary 0.3.3 (Triangle inequality)**.**
For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$,

$$\|\mathbf{u} + \mathbf{v}\|_2 \leq \|\mathbf{u}\|_2 + \|\mathbf{v}\|_2 .$$

Proof. By Cauchy–Schwarz,

$$\|\mathbf{u} + \mathbf{v}\|_2^2 \leq \left(\|\mathbf{u}\|_2 + \|\mathbf{v}\|_2\right)^2.$$

Take the non-negative square root. □

0.4. Shadows and projections

Definition 0.4.1 (Projection onto a direction)**.**
Let $\mathbf{e}$ be a unit vector.

The scalar shadow of $\mathbf{v}$ on $\mathbf{e}$ is

$$\mathbf{e} \cdot \mathbf{v}.$$

The vector shadow, or projection, is

$$\boxed{P_{\mathbf{e}}\mathbf{v} = (\mathbf{e} \cdot \mathbf{v})\mathbf{e}}.$$

Worked example 0.4.2 (A shadow on a line)**.**
Let

$$\mathbf{e} = \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j}), \qquad \mathbf{v} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}.$$

Then

$$\mathbf{e} \cdot \mathbf{v} = 2\sqrt{2}, \qquad P_{\mathbf{e}}\mathbf{v} = 2\mathbf{i} + 2\mathbf{j}.$$

The remainder $\mathbf{v} - P_{\mathbf{e}}\mathbf{v}$ is perpendicular to $\mathbf{e}$.

Motivational bridge

The familiar formula

$$(\mathbf{e} \cdot \mathbf{v})\mathbf{e}$$

becomes $\langle e, v \rangle\, e$ in an abstract inner-product space. It is the same "shadow" in every part of these notes.

Part I.

Finite-Dimensional Vectors over the Real Field

1. Finite-Dimensional Vectors: $\mathbb{R}^N$

Part I. The finite-dimensional model: finite sums, matrices and no convergence questions.

1.1. Vectors and vector spaces

Definition 1.1.1 (Real vector space)**.**
A real vector space is a set V with addition and scalar multiplication by $\mathbb{R}$, satisfying the vector-space axioms.

The model example is

$$\mathbb{R}^N = \{(v_1, \ldots, v_N) : v_i \in \mathbb{R}\}.$$

Here v_i is the ith component. Vectors may represent arrows, coefficients, data or discretised fields.

Worked example 1.1.2 (A force and a coefficient vector)**.**
$F = (3, -4, 0) \in \mathbb{R}^3$ may be a force. The vector $c = (1, -2, 5, 7) \in \mathbb{R}^4$ may store the coefficients of a cubic polynomial.

Motivational bridge

The index set is presently finite:

$$v = (v_1, \ldots, v_N), \qquad i \in \{1, \ldots, N\}.$$

Parts II and III change only the index set.

1.2. Linear combinations, span and linear independence

Definition 1.2.1.
A *linear combination* of $v_1, \ldots, v_k \in V$ has the form $\sum_{j=1}^{k} \alpha_j v_j$. Their span is the set of these combinations. They are linearly independent if

$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_k = 0.$$

Span records what the vectors generate; independence removes redundancy.

Worked example 1.2.2 (Testing span and independence)**.**
Let $u = (1, 1, 0)$ and $v = (1, -1, 0)$. If $\alpha u + \beta v = 0$, then $(\alpha+\beta, \alpha-\beta, 0) = 0$, so $\alpha = \beta = 0$. Thus u, v are independent. Their span is the plane $\{(x, y, 0) : x, y \in \mathbb{R}\}$, since

$$(x, y, 0) = \frac{x + y}{2}u + \frac{x - y}{2}v.$$

1.3. Bases and coordinates

Definition 1.3.1.
A basis of V is a linearly independent set that spans V. The standard basis of $\mathbb{R}^N$ is $e_1, \ldots, e_N$, where $(e_i)_j = \delta_{ij}$ and

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

A basis supplies coordinates. The vector is fixed; its coordinates depend on the chosen basis.

Worked example 1.3.2 (The same vector in two bases)**.**
In $\mathbb{R}^2$ let $b_1 = (1, 1)$ and $b_2 = (1, -1)$. Since

$$(3, 1) = 2b_1 + b_2,$$

the standard coordinates of the vector are $(3, 1)$, whereas its coordi-

nates in the ordered basis (b_1, b_2) are $(2, 1)$.

1.4. Decomposition in a basis

Proposition 1.4.1 (Unique decomposition)**.**
If $(b_1, \ldots, b_N)$ is a basis of V, then every $v \in V$ has a unique representation

$$\boxed{v = \sum_{i=1}^{N} v_i b_i}.$$

Proof. Existence follows because the basis spans V.

For uniqueness, suppose

$$v = \sum_{i=1}^{N} v_i b_i \qquad \text{and} \qquad v = \sum_{i=1}^{N} w_i b_i.$$

Subtracting gives

$$\sum_{i=1}^{N} (v_i - w_i) b_i = 0.$$

Linear independence gives

$$v_i = w_i \qquad (i = 1, \ldots, N).$$

□

Worked example 1.4.2 (A polynomial as a coordinate vector)**.**
In the vector space of polynomials of degree at most two, take the basis $(1, x, x^2)$. Then

$$p(x) = 2 - 3x + 5x^2$$

has coordinate vector $(2, -3, 5)$. With the alternative basis $(1, x, 1+x^2)$ it becomes $p(x) = -3 + (-3)x + 5(1 + x^2)$, so its coordinate vector is $(-3, -3, 5)$.

1.5. Linear operators

Definition 1.5.1.
A map $A : V \to W$ is linear if

$$A(\alpha u + \beta v) = \alpha Au + \beta Av$$

for all $u, v \in V$ and $\alpha, \beta \in \mathbb{R}$.

Linearity is the superposition principle.

Worked example 1.5.2 (Rotation through a right angle)**.**
The operator $R : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $R(x, y) = (-y, x)$ is linear. It is

represented in the standard basis by

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

By contrast, $T(x, y) = (x + 1, y)$ is not linear because $T(0, 0) \neq (0, 0)$.

1.6. Matrix elements and operator action

Let

$$A : V \longrightarrow W$$

be a linear operator. It takes an input vector $v \in V$ to an output vector $Av \in W$.

Choose a basis (b_j) for the input space V and a basis (c_i) for the output space W.

The numbers A_{ij} are defined by

$$\boxed{Ab_j = \sum_i A_{ij} c_i}.$$

Thus the jth column records the coordinates of Ab_j.

Now let

$$v = \sum_j v_j b_j.$$

Linearity gives

$$Av = A\left(\sum_j v_j b_j\right) = \sum_j v_j A b_j.$$

Substitute the basis expansion of Ab_j:

$$Av = \sum_i \left(\sum_j A_{ij} v_j\right) c_i.$$

Therefore the ith output component is

$$\boxed{(Av)_i = \sum_{j=1}^{N} A_{ij} v_j}.$$

The row label i selects an output component. The column label j runs over the input components.

Worked example 1.6.1 (Reading rows as output components)**.** For

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}, \qquad v = \begin{pmatrix} 5 \\ 2 \end{pmatrix},$$

one obtains $(Av)_1 = 2(5) - 1(2) = 8$ and $(Av)_2 = 3(5) + 4(2) = 23$. Thus $Av = (8, 23)^T$.

Motivational bridge

Keep the component formula visible. It will become

$$(Av)_i = \sum_j A_{ij} v_j \quad \longrightarrow \quad (Aa)_n = \sum_m A_{nm} a_m,$$

$$(Af)_x = \sum_y (A_{xy} f_y) \quad \rightsquigarrow \quad \int A(x,y) f(y)\, \mathrm{d}y.$$

1.7. Inner products

Definition 1.7.1.
An inner product is a bilinear, symmetric, positive-definite map $V \times V \to \mathbb{R}$. On $\mathbb{R}^N$ the standard choice is

$$\boxed{\langle u, v \rangle = \sum_{i=1}^{N} u_i v_i = u^T v}.$$

The proof in Section 0.3 uses only the inner-product rules. It therefore gives Cauchy–Schwarz in $\mathbb{R}^N$:

$$|\langle u, v \rangle| \leq \|u\|_2 \, \|v\|_2 .$$

For non-zero vectors, $\cos\theta = \langle u, v \rangle / (\|u\|_2 \, \|v\|_2)$.

Worked example 1.7.2 (Angle from an inner product)**.**
For $u = (1, 1, 0)$ and $v = (1, 0, 1)$,

$$\langle u, v \rangle = 1, \qquad \|u\|_2 = \|v\|_2 = \sqrt{2}, \qquad \cos\theta = \frac{1}{2}.$$

Hence $\theta = \pi/3$.

1.8. Norms and p-norms

The abstract definition of a norm was given in Section 0.2.

Definition 1.8.1 (p-norms on $\mathbb{R}^N$)**.**
For $1 \le p < \infty$,

$$\|v\|_p = \left(\sum_{i=1}^{N} |v_i|^p \right)^{1/p}, \qquad \|v\|_\infty = \max_{1 \le i \le N} |v_i|.$$

The inner-product norm is $\|v\|_2 = \sqrt{\langle v, v \rangle}$.

Worked example 1.8.2 (Different measurements of the same vector)**.**
For $v = (3, -4)$,

$$\|v\|_1 = 7, \qquad \|v\|_2 = 5, \qquad \|v\|_\infty = 4.$$

The values differ; the induced notions of convergence do not.

1.9. Normalisation and orthogonality

Definition 1.9.1.
A non-zero vector v is normalised by setting

$$\widehat{v} = \frac{v}{\|v\|_2}.$$

Vectors u and v are orthogonal if $\langle u, v\rangle = 0$.

Worked example 1.9.2 (Normalising an orthogonal pair)**.**
The vectors $u = (1, 1)$ and $v = (1, -1)$ are orthogonal. Their normalised forms are

$$\widehat{u} = \frac{1}{\sqrt{2}}(1, 1), \qquad \widehat{v} = \frac{1}{\sqrt{2}}(1, -1).$$

The Pythagorean identity follows immediately: $\|\alpha\widehat{u} + \beta\widehat{v}\|_2^2 = \alpha^2 + \beta^2$.

1.10. Orthonormal bases and projections

Definition 1.10.1.
A basis $(e_1, \ldots, e_N)$ is orthonormal if $\langle e_i, e_j \rangle = \delta_{ij}$. The orthogonal projection onto the line $\operatorname{span}\{e_i\}$ is

$$\boxed{P_i v = \langle e_i, v \rangle \, e_i}.$$

More generally, if $U = \operatorname{span}\{e_1, \ldots, e_k\}$, then $P_U v = \sum_{i=1}^{k} \langle e_i, v \rangle \, e_i$.

Geometrically, $P_U v$ is the shadow of v on U.

The remainder $v - P_U v$ points away from the subspace and satisfies

$$v - P_U v \perp U.$$

Also $P_U^2 = P_U$: projecting a shadow again does not change it.

Worked example 1.10.2 (Nearest point on a line)**.**
Let $e = \frac{1}{\sqrt{2}}(1, 1)$ and $v = (3, 1)$. Then

$$P_e v = \langle e, v \rangle \, e = \frac{4}{\sqrt{2}} \frac{1}{\sqrt{2}} (1, 1) = (2, 2).$$

The error $(1, -1)$ is orthogonal to the line. Thus $(2, 2)$ is also the point on the line nearest to $(3, 1)$.

1.11. Equivalence of norms in finite dimensions

Theorem 1.11.1 (p-norm equivalence on $\mathbb{R}^N$)**.**
Let $1 \le p, q \le \infty$.

There is a constant $C \ge 1$*, independent of* v*, such that*

$$\boxed{C^{-1} \|v\|_p \le \|v\|_q \le C \|v\|_p}$$

for every $v \in \mathbb{R}^N$.

Hence all p*-norms on a fixed* $\mathbb{R}^N$ *are equivalent.*

A proof is given in Appendix A.

Worked example 1.11.2 (Why the constant depends on dimension)**.**
For $\mathbf{1}_N = (1, \dots, 1)$, $\|\mathbf{1}_N\|_p = N^{1/p}$ and $\|\mathbf{1}_N\|_q = N^{1/q}$. The ratio therefore changes with N.

This causes no problem while N is fixed. It becomes decisive in Part II.

Corollary 1.11.3.
A sequence of vectors in $\mathbb{R}^N$ *converges in one* p*-norm if and only if it converges in every other* p*-norm.*

1.12. Resolution of the identity and Parseval's identity

Proposition 1.12.1 (Basis expansion)**.**
Let $(e_i)_{i=1}^N$ *be an orthonormal basis.*

Every vector satisfies

$$v = \sum_{i=1}^{N} \langle e_i, v \rangle \, e_i .$$

Equivalently,

$$\boxed{\mathrm{I} = \sum_{i=1}^{N} P_i} .$$

This is the resolution of the identity.

Proposition 1.12.2 (Parseval's identity)**.**
For every $v \in \mathbb{R}^N$*,*

$$\boxed{\|v\|_2^2 = \sum_{i=1}^{N} |\, \langle e_i, v \rangle \,|^2} .$$

Worked example 1.12.3 (Resolving a vector)**.**
For $e_\pm = 2^{-1/2}(1, \pm 1)$ and $v = (3, 1)$,

$$P_+ v = (2, 2), \qquad P_- v = (1, -1), \qquad v = P_+ v + P_- v.$$

Also $\|v\|_2^2 = 10 = |\langle e_+, v\rangle|^2 + |\langle e_-, v\rangle|^2 = 8 + 2$.

1.13. Eigenvalues and eigenvectors

Definition 1.13.1.
A non-zero vector v is an eigenvector of A with eigenvalue $\lambda \in \mathbb{R}$ if

$$\boxed{Av = \lambda v}.$$

Equivalently, $(A - \lambda \mathrm{I})v = 0$. For a matrix, eigenvalues satisfy $\det(A - \lambda I) = 0$.

Worked example 1.13.2 (A coupled two-component operator)**.**
Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its characteristic polynomial is $(2-\lambda)^2 - 1$, giving eigenvalues 3 and 1. Corresponding normalised eigenvectors are $e_+ = 2^{-1/2}(1, 1)$ and $e_- = 2^{-1/2}(1, -1)$. Along these special directions, A acts only by stretching.

1.14. Eigenspaces and degeneracy

Definition 1.14.1.
The eigenspace belonging to λ is

$$E_\lambda = \ker(A - \lambda \mathrm{I}).$$

Its dimension is the geometric multiplicity. An eigenvalue is called degenerate when its eigenspace has dimension greater than one.

Worked example 1.14.2 (A degenerate eigenvalue)**.**
For $A = \mathrm{diag}(2, 2, 5)$, the eigenvalue 2 has eigenspace

$$E_2 = \mathrm{span}\{(1, 0, 0), (0, 1, 0)\},$$

so it is doubly degenerate. Every non-zero vector in the xy-plane is an eigenvector with eigenvalue 2.

1.15. Adjoints and self-adjoint operators

Definition 1.15.1.
The adjoint A^* is characterised by

$$\boxed{\langle u, Av \rangle = \langle A^* u, v \rangle}$$

for all u, v. In a real orthonormal basis, $A^* = A^T$. The operator is self-adjoint if $A^* = A$; equivalently, its matrix is symmetric.

Worked example 1.15.2 (Adjoint versus self-adjoint)**.**
If $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$, then $A^* = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 3 \end{smallmatrix}\right)$, so A is not self-adjoint. The matrix $B = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$ is self-adjoint.

1.16. Real eigenvalues of self-adjoint operators

The following inner-product calculation will be reused in every part.

Proposition 1.16.1.
If A is self-adjoint and $Av = \lambda v$ for $v \neq 0$, then the eigenvalue is obtained from the real Rayleigh quotient

$$\boxed{\lambda = \frac{\langle v, Av \rangle}{\langle v, v \rangle}}.$$

Proof. Taking the inner product of $Av = \lambda v$ with v gives $\langle v, Av \rangle = \lambda \langle v, v \rangle$. Since $A = A^*$, $\langle v, Av \rangle = \langle Av, v \rangle$ is a real scalar, and $\langle v, v \rangle > 0$. □

Worked example 1.16.2 (Rayleigh quotient)**.**

For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $v = (1, 1)$, the quotient is $\lambda = (v^T A v)/(v^T v) = 6/2 = 3$, recovering the eigenvalue without a determinant.

1.17. Orthogonality of eigenvectors

Theorem 1.17.1.
Eigenvectors of a self-adjoint operator belonging to distinct eigenvalues are orthogonal.

Proof. Suppose $Au = \lambda u$ and $Av = \mu v$. Self-adjointness gives

$$\mu \langle u, v \rangle = \langle u, Av \rangle = \langle Au, v \rangle = \lambda \langle u, v \rangle .$$

Thus $(\mu - \lambda) \langle u, v \rangle = 0$. If $\lambda \neq \mu$, then $\langle u, v \rangle = 0$. □

Within a degenerate eigenspace, Gram–Schmidt produces an orthonormal basis.

Worked example 1.17.2.
For the matrix in Section 1.13, the eigenvectors $(1, 1)$ and $(1, -1)$ belong to 3 and 1, and their inner product is zero exactly as the theorem predicts.

1.18. The finite-dimensional spectral theorem

Theorem 1.18.1 (Spectral theorem over $\mathbb{R}$)**.**
Let A be self-adjoint on a finite-dimensional real inner-product space.

Then:

(i) every eigenvalue of A is real;

(ii) eigenspaces for distinct eigenvalues are orthogonal;

(iii) the space has an orthonormal basis of eigenvectors of A;

(iv) in this basis, A has the form

$$A = QDQ^T, \qquad Q^TQ = I,$$

where Q is orthogonal and D is real diagonal.

Thus a self-adjoint operator has enough orthogonal eigenvectors to resolve every vector.

Worked example 1.18.2 (Orthogonal diagonalisation)**.**
For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, take

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $Q^TQ = I$ and $A = QDQ^T$.

1.19. Spectral decomposition

Let P_λ be the orthogonal projection onto E_λ. The spectral theorem gives

$$\boxed{A = \sum_\lambda \lambda P_\lambda}, \qquad \boxed{\mathrm{I} = \sum_\lambda P_\lambda}.$$

If every eigenspace is one-dimensional with normalised eigenvector e_i, then $P_i v = \langle e_i, v\rangle\, e_i$ and $A = \sum_i \lambda_i P_i$.

The operator only multiplies coefficients

Let $q_1, \dots, q_N$ be an orthonormal eigenbasis and write

$$v = c_1 q_1 + \cdots + c_N q_N, \qquad c_i = \langle q_i, v\rangle\,.$$

Since $A q_i = \lambda_i q_i$,

$$\boxed{A v = \lambda_1 c_1 q_1 + \cdots + \lambda_N c_N q_N.}$$

The eigenvectors do not change direction. The ith coordinate is simply multiplied by λ_i:

$$c_i \quad \longmapsto \quad \lambda_i c_i.$$

Motivational bridge

Let P be the matrix whose columns are the eigenvectors, and let

$$E = \operatorname{diag}(\lambda_1, \dots, \lambda_N).$$

Then

$$\boxed{A = PEP^{-1}}.$$

The three actions are

$$v \xmapsto{P^{-1}} c \xmapsto{E} (\lambda_i c_i)_i \xmapsto{P} Av.$$

For a real self-adjoint operator, P is orthogonal, so $P^{-1} = P^T$. Choosing the eigenbasis turns the operator into coefficient multiplication.

Because $P_i = q_i q_i^T$,

$$\boxed{A = \sum_{i=1}^{N} \lambda_i q_i q_i^T = \sum_{i=1}^{N} \lambda_i P_i.}$$

Each term extracts one eigendirection, multiplies it by its eigenvalue, and puts it back.

Worked example 1.19.1 (One matrix, two coordinate multipliers)**.** For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad q_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad q_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

the eigenvalues are 3 and 1. If $v = (a, b)^T$, then

$$c_+ = \frac{a+b}{\sqrt{2}}, \qquad c_- = \frac{a-b}{\sqrt{2}},$$

and therefore

$$\boxed{Av = 3c_+q_+ + c_-q_-.}$$

In the eigenbasis, the entire matrix calculation is $(c_+, c_-) \mapsto (3c_+, c_-)$.

Worked example 1.19.2 (Using the decomposition)**.**
For the matrix above,

$$P_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad P_- = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

so $A = 3P_+ + P_-$. Consequently

$$A^k = 3^k P_+ + P_-, \qquad \mathrm{e}^{tA} = \mathrm{e}^{3t}P_+ + \mathrm{e}^{t}P_-.$$

Operator functions reduce to scalar functions of the eigenvalues.

$$v_i \longrightarrow \langle e_i, v\rangle \longrightarrow P_i v \longrightarrow \mathrm{I} = \sum_i P_i \longrightarrow A = \sum_\lambda \lambda P_\lambda.$$

Part II.

Infinite-Dimensional Vectors with a Natural-Number Index

2. Infinite-Dimensional Vectors (Natural): ℓ^2

Part II. Coordinates are indexed by $n \in \mathbb{N}$. Infinite sums require convergence; operators require boundedness or a stated domain.

Motivational bridge

$$(v_1, \ldots, v_N) \quad \longrightarrow \quad (a_1, a_2, a_3, \ldots), \qquad \sum_{i=1}^{N} \quad \longrightarrow \quad \sum_{n=1}^{\infty}.$$

The algebra is unchanged. Convergence is new.

2.1. Vectors and sequence spaces

Definition 2.1.1 (ℓ^p spaces)**.**
For $1 \le p < \infty$,

$$\ell^p = \left\{ a = (a_n)_{n \ge 1} : a_n \in \mathbb{R},\ \sum_{n=1}^{\infty} |a_n|^p < \infty \right\},$$

and ℓ^∞ consists of bounded real sequences. The space central to this part is

$$\boxed{\ell^2 = \left\{ a = (a_n) : \sum_{n=1}^{\infty} a_n^2 < \infty \right\}.}$$

Operations are componentwise. Square summability makes $\|a\|_2$ finite.

Worked example 2.1.2 (Membership matters)**.**
The sequence $a_n = 1/n$ lies in ℓ^2 because $\sum_n 1/n^2 < \infty$, but it does not lie in ℓ^1 because the harmonic series diverges. The constant sequence $(1, 1, \ldots)$ lies in neither ℓ^1 nor ℓ^2, although it lies in ℓ^∞.

2.2. Linear combinations, span and linear independence

The definitions from Section 1.2 are unchanged. Infinite expansions are limits, not algebraic combinations.

Let c_{00} denote the finitely supported sequences. Then

$$c_{00} = \operatorname{span}\{e_1, e_2, \ldots\}, \qquad \overline{c_{00}} = \ell^2.$$

Worked example 2.2.1 (Approximating by finite vectors)**.**

For $a = (1, 1/2, 1/3, \ldots)$, define

$$a^{(N)} = (1, 1/2, \ldots, 1/N, 0, 0, \ldots).$$

Then

$$\left\| a - a^{(N)} \right\|_2^2 = \sum_{n>N} \frac{1}{n^2} \longrightarrow 0.$$

Thus a genuinely infinite vector is the limit of finite linear combinations of the e_n.

2.3. Bases and coordinates

The standard sequence e_n has a 1 in position n and zeros elsewhere. Then $\langle e_n, e_m \rangle = \delta_{nm}$. In Hilbert-space language, $(e_n)_{n\geq 1}$ is a complete orthonormal system, or Hilbert basis, for ℓ^2.

Important distinction

Here an orthonormal basis gives a norm-convergent series expansion. A Hamel basis uses finite combinations.

Worked example 2.3.1 (Coordinates are inner products)**.**
For $a = (a_1, a_2, \ldots) \in \ell^2$,

$$\langle e_n, a \rangle = a_n.$$

The familiar coordinate-extraction formula survives unchanged.

2.4. Decomposition and convergence

Every $a \in \ell^2$ has the norm-convergent expansion

$$\boxed{a = \sum_{n=1}^{\infty} a_n e_n}.$$

This equation means

$$\left\| a - \sum_{n=1}^{N} a_n e_n \right\|_2^2 = \sum_{n>N} |a_n|^2 \longrightarrow 0.$$

Definition 2.4.1 (Completeness)**.**
A normed space is complete if every Cauchy sequence of its vectors converges to a vector in the space. A complete inner-product space is called a Hilbert space. The space ℓ^2 is a Hilbert space.

Completeness keeps Cauchy limits inside the space.

Worked example 2.4.2 (A controlled truncation error)**.**

For $a_n = 2^{-n}$,

$$\left\| a - \sum_{n=1}^{N} 2^{-n} e_n \right\|_2^2 = \sum_{n>N} 4^{-n} = \frac{4^{-N}}{3}.$$

The norm error is therefore $2^{-N}/\sqrt{3}$.

2.5. Linear operators

Linearity still means $A(\alpha a + \beta b) = \alpha Aa + \beta Ab$. In infinite dimensions we must also ask where A is defined and whether it is continuous.

Definition 2.5.1 (Bounded operator)**.**
A linear operator $A : \ell^2 \to \ell^2$ is bounded if there is $C < \infty$ such that

$$\|Aa\|_2 \le C \, \|a\|_2$$

for every $a \in \ell^2$. Its operator norm is $\|A\| = \sup_{\|a\|_2 = 1} \|Aa\|_2$.

Bounded linear operators are continuous. An unbounded operator must be given with a domain $\mathcal{D}(A) \subsetneq \ell^2$.

Worked example 2.5.2 (Shift and an unbounded diagonal operator)**.**

The right shift

$$S(a_1, a_2, \ldots) = (0, a_1, a_2, \ldots)$$

satisfies $\|Sa\|_2 = \|a\|_2$, so $\|S\| = 1$. By contrast, $D(a_1, a_2, \ldots) = (a_1, 2a_2, 3a_3, \ldots)$ is not defined on all of ℓ^2. Its natural domain is $\mathcal{D}(D) = \{a : \sum_n n^2 a_n^2 < \infty\}$.

2.6. Infinite matrices and operator action

Define $A_{nm} = \langle e_n, Ae_m \rangle$ whenever this is meaningful. The matrix rule becomes

$$\boxed{(Aa)_n = \sum_{m=1}^{\infty} A_{nm} a_m}.$$

The series must converge and its output must lie in ℓ^2. Not every array (A_{nm}) defines an operator.

Worked example 2.6.1 (Diagonal and shift matrices)**.**
For $(Aa)_n = \lambda_n a_n$, the infinite matrix is diagonal: $A_{nm} = \lambda_n \delta_{nm}$. It is bounded precisely when $\sup_n |\lambda_n| < \infty$, and then $\|A\| = \sup_n |\lambda_n|$. The right shift has entries $S_{nm} = \delta_{n,m+1}$.

Motivational bridge

The column index now runs forever:

$$(Av)_i = \sum_{j=1}^{N} A_{ij} v_j \quad \longrightarrow \quad (Aa)_n = \sum_{m=1}^{\infty} A_{nm} a_m.$$

Convergence is essential.

2.7. Inner products

Definition 2.7.1.
For $a, b \in \ell^2$, define

$$\boxed{\langle a, b \rangle = \sum_{n=1}^{\infty} a_n b_n}.$$

The series converges absolutely by Cauchy–Schwarz: $\sum_n |a_n b_n| \le \|a\|_2 \, \|b\|_2$.

Worked example 2.7.2 (A geometric inner product)**.**
Let $a_n = 2^{-n}$ and $b_n = (-1)^{n-1} 2^{-n}$. Then

$$\langle a, b \rangle = \sum_{n=1}^{\infty} (-1)^{n-1} 4^{-n} = \frac{1/4}{1 + 1/4} = \frac{1}{5}.$$

Also $\|a\|_2^2 = \|b\|_2^2 = \sum_n 4^{-n} = 1/3$.

2.8. Norms and p-norms

For $1 \le p < \infty$,

$$\|a\|_p = \left(\sum_{n=1}^{\infty} |a_n|^p \right)^{1/p}, \qquad \|a\|_\infty = \sup_n |a_n|.$$

Each norm has its natural space ℓ^p. On ℓ^2, $\|a\|_2 = \sqrt{\langle a, a \rangle}$.

Worked example 2.8.1 (Power-law tails)**.**
For $a_n = n^{-\alpha}$,

$$a \in \ell^p \quad \Longleftrightarrow \quad \alpha p > 1.$$

Thus $a_n = n^{-3/4}$ lies in ℓ^2 but not ℓ^1. The value of p changes the admissible sequences.

2.9. Normalisation and orthogonality

For non-zero $a \in \ell^2$, set $\widehat{a} = a/\|a\|_2$. Two sequences are orthogonal if $\sum_n a_n b_n = 0$.

Worked example 2.9.1 (Normalising a geometric sequence)**.**
Let $a = (1, 1/2, 1/4, \ldots)$, with the first term indexed by $n = 0$ for this

example. Then

$$\|a\|_2^2 = \sum_{n=0}^{\infty} 4^{-n} = \frac{4}{3}, \qquad \widehat{a} = \frac{\sqrt{3}}{2}(1, 1/2, 1/4, \ldots).$$

The sequences $u = (1, 1, 0, \ldots)$ and $v = (1, -1, 0, \ldots)$ are orthogonal just as their two-dimensional versions were.

2.10. Orthonormal bases and projections

For the standard orthonormal basis (e_n),

$$P_n a = \langle e_n, a \rangle \, e_n = a_n e_n.$$

More generally, for a normalised $u \in \ell^2$, $P_u a = \langle u, a \rangle \, u$. This is again a bounded, self-adjoint idempotent: $P_u^2 = P_u = P_u^*$.

Worked example 2.10.1 (Projecting onto a two-dimensional model)**.** Let $U = \operatorname{span}\{e_1, e_2\}$. Then

$$P_U(a_1, a_2, a_3, \ldots) = (a_1, a_2, 0, 0, \ldots).$$

The error has norm $(\sum_{n \geq 3} |a_n|^2)^{1/2}$ and is orthogonal to every vector in U. Finite-mode approximation is therefore an orthogonal projection.

2.11. Failure of equivalence of norms in infinite dimensions

Theorem 2.11.1 (No dimension-independent equivalence)**.**
The p-norms are not equivalent on infinite sequence spaces. In particular, there are no constants $c, C > 0$ such that

$$c\,\|a\|_1 \le \|a\|_2 \le C\,\|a\|_1$$

on a common infinite-dimensional space in both directions, and the spaces ℓ^p themselves need not contain the same sequences.

Proof. For each N, let

$$u^{(N)} = (\underbrace{1, \ldots, 1}_{N \text{ terms}}, 0, 0, \ldots).$$

Then $\left\|u^{(N)}\right\|_p = N^{1/p}$ and $\left\|u^{(N)}\right\|_q = N^{1/q}$. If $p < q$, the ratio

$$\frac{\left\|u^{(N)}\right\|_p}{\left\|u^{(N)}\right\|_q} = N^{1/p - 1/q} \longrightarrow \infty.$$

No constant independent of N can compare these norms in the way available on a fixed $\mathbb{R}^N$. □

Worked example 2.11.2 (The vector sets also differ)**.**
The sequence $(1/n)$ is in ℓ^2 but not ℓ^1. Therefore it is not even meaningful to say that the ℓ^1 and ℓ^2 norms are two equivalent measurements of exactly the same infinite-dimensional space.

Important distinction

In Section 1.11, N was fixed. Here $N^{1/p-1/q}$ is unbounded. Norm choice can change convergence and membership.

2.12. Resolution of the identity and Parseval's identity

Proposition 2.12.1 (Basis expansion)**.**
For each $a \in \ell^2$,

$$a = \lim_{N\to\infty} \sum_{n=1}^{N} P_n a.$$

Equivalently,

$$\boxed{\mathrm{I} = \sum_{n=1}^{\infty} P_n},$$

meaning that the partial sums converge after acting on each fixed vector.

Proposition 2.12.2 (Parseval's identity)**.**
For every $a \in \ell^2$,

$$\boxed{\|a\|_2^2 = \sum_{n=1}^{\infty} |\langle e_n, a\rangle|^2}.$$

Worked example 2.12.3 (Energy captured by the first modes)**.**

If $a_n = 1/n$, then the first N coordinate projections capture energy $\sum_{n=1}^{N} 1/n^2$. The missing energy is the tail $\sum_{n>N} 1/n^2$, which tends to zero. In applications this is the basic logic behind modal truncation.

2.13. Eigenvalues and eigenvectors

The equation is unchanged:

$$Aa = \lambda a,$$

A solution must lie in both $\mathcal{D}(A)$ and ℓ^2.

Worked example 2.13.1 (Diagonal eigenvectors)**.**
For $(Aa)_n = \lambda_n a_n$, each standard basis vector satisfies $Ae_n = \lambda_n e_n$. If $\lambda_n \to 0$, these eigenvalues may accumulate at 0, a phenomenon impossible for a finite matrix with only finitely many eigenvalues.

Worked example 2.13.2 (An operator with no eigenvectors)**.**
The right shift $S(a_1, a_2, \ldots) = (0, a_1, a_2, \ldots)$ has no non-zero eigenvector over ℓ^2. Indeed, $Sa = \lambda a$ forces either $a_1 = a_2 = \cdots = 0$ recursively or, when $\lambda = 0$, again $a = 0$. An infinite-dimensional operator need not possess even one eigenvector.

2.14. Eigenspaces and degeneracy

As before, $E_\lambda = \ker(A - \lambda \mathrm{I})$. Its dimension may be finite or infinite.

Worked example 2.14.1 (Infinite degeneracy)**.**
Define A by

$$A(a_1, a_2, a_3, a_4, \ldots) = (a_1, 2a_2, a_3, 2a_4, \ldots).$$

The eigenvalue 1 has eigenspace generated by $e_1, e_3, e_5, \ldots$, while 2 has eigenspace generated by $e_2, e_4, e_6, \ldots$. Both eigenvalues have infinite degeneracy.

2.15. Adjoints and self-adjoint operators

For bounded A, the adjoint satisfies

$$\boxed{\langle a, Ab\rangle = \langle A^* a, b\rangle}.$$

For unbounded operators, the adjoint domain is part of the definition.

Worked example 2.15.1 (Adjoint shifts)**.**

For the right shift S,

$$\langle a, Sb\rangle = a_2b_1 + a_3b_2 + \cdots = \langle (a_2, a_3, \ldots), b\rangle .$$

Hence S^* is the left shift $S^*(a_1, a_2, \ldots) = (a_2, a_3, \ldots)$. Thus $S \neq S^*$. A bounded real diagonal operator with real entries is self-adjoint.

2.16. Real eigenvalues of self-adjoint operators

Proposition 2.16.1.
If A is self-adjoint and $Aa = \lambda a$ for $a \neq 0$, then

$$\lambda = \frac{\langle a, Aa\rangle}{\langle a, a\rangle} \in \mathbb{R}.$$

Proof. The proof is identical to Section 1.16: $\langle a, Aa\rangle = \lambda \langle a, a\rangle$, while self-adjointness makes the quadratic form real. For an unbounded operator, the calculation is valid for $a \in \mathcal{D}(A)$. □

Worked example 2.16.2 (Diagonal Rayleigh quotient)**.**
For $(Aa)_n = \lambda_n a_n$ with bounded real (λ_n),

$$\frac{\langle a, Aa\rangle}{\langle a, a\rangle} = \frac{\sum_n \lambda_n a_n^2}{\sum_n a_n^2}.$$

This is a weighted average of the diagonal values, with non-negative weights $a_n^2/\|a\|_2^2$.

2.17. Orthogonality of eigenvectors

Theorem 2.17.1.
Eigenvectors of a self-adjoint operator belonging to distinct eigenvalues are orthogonal.

Proof. If $Aa = \lambda a$ and $Ab = \mu b$, then

$$\mu \langle a, b\rangle = \langle a, Ab\rangle = \langle Aa, b\rangle = \lambda \langle a, b\rangle .$$

When $\lambda \neq \mu$, it follows that $\langle a, b\rangle = 0$. □

The new question is whether the eigenvectors span the space.

Worked example 2.17.2.
For Section 2.14's diagonal operator, the odd and even eigenspaces are orthogonal. Their products vanish coordinate by coordinate.

2.18. Spectral theorem and spectral issues

An orthonormal eigenbasis need not exist.

Theorem 2.18.1 (Compact self-adjoint case, informal form)**.**

Let A be compact and self-adjoint on a real Hilbert space.

Then:

(i) every non-zero spectral value is a real eigenvalue;

(ii) each non-zero eigenspace is finite-dimensional;

(iii) non-zero eigenvalues can accumulate only at 0*;*

(iv) the corresponding eigenvectors, together with an orthonormal basis of ker *A, form an orthonormal basis of the space.*

General self-adjoint operators require spectral projections. The spectrum replaces the finite eigenvalue list.

Worked example 2.18.2 (A compact diagonal operator)**.**
The operator $(Aa)_n = a_n/n$ is compact and self-adjoint. Its eigenvalues are $1, 1/2, 1/3, \ldots$, with eigenvectors e_n, and the only accumulation point is 0.

Important distinction

Self-adjointness gives a real spectrum, not necessarily a countable eigenbasis. Part III gives a continuous-spectrum example.

2.19. Spectral decomposition

When a self-adjoint operator has a complete orthonormal eigenbasis (e_n),

$$\boxed{A = \sum_{n=1}^{\infty} \lambda_n P_n}, \qquad P_n a = \langle e_n, a \rangle \, e_n,$$

with the series interpreted through its action on vectors. More generally, the spectral theorem takes the form

$$\boxed{A = \int_{\sigma(A)} \lambda \, \mathrm{d}P(\lambda)},$$

where P is a family of orthogonal spectral projections. The sum is the discrete special case.

Worked example 2.19.1 (Functions of a diagonal operator)**.**
For $(Aa)_n = n^{-1} a_n$,

$$A^k a = \sum_{n=1}^{\infty} n^{-k} P_n a, \qquad \mathrm{e}^{tA} a = \sum_{n=1}^{\infty} \mathrm{e}^{t/n} P_n a.$$

Again, operator functions reduce to scalar functions of spectral values.

$$a_n \longrightarrow \langle e_n, a\rangle \longrightarrow P_n a \longrightarrow \mathrm{I} = \sum_{n=1}^{\infty} P_n \longrightarrow A = \sum_n \lambda_n P_n \text{ when an eigenbasis exists.}$$

Part III.

Infinite-Dimensional Vectors with a Real-Number Index

3. Infinite-Dimensional Vectors (Real): L^2

Part III. A function is a continuously indexed vector. Write f_x at first; then return to $f(x)$. Sums become integrals and matrices become kernels.

Motivational bridge

Think of a function as a continuously indexed vector:

$$a = (a_n)_{n\in\mathbb{N}} \quad \longrightarrow \quad f = (f_x)_{x\in I}.$$

Temporarily write

$$\sum_x (\cdots) \quad \rightsquigarrow \quad \int_I (\cdots)\,\mathrm{d}x.$$

The symbol $\sum_x$ is heuristic only. It is not a sum over uncountably many points; $\int_I \mathrm{d}x$ is the rigorous object.

3.1. Vectors and function spaces

Initially write the component at x as f_x. Operations are componentwise:

$$(f+g)_x = f_x + g_x, \qquad (\alpha f)_x = \alpha f_x.$$

In standard notation: $(f+g)(x) = f(x) + g(x)$ and $(\alpha f)(x) = \alpha f(x)$.

Definition 3.1.1 (L^p spaces)**.**
For an interval I and $1 \le p < \infty$,

$$L^p(I) = \left\{ f : I \to \mathbb{R} : \int_I |f(x)|^p \, \mathrm{d}x < \infty \right\},$$

Functions equal almost everywhere are identified. The central space is

$$\boxed{L^2(I) = \left\{ f : \int_I |f(x)|^2 \, \mathrm{d}x < \infty \right\}}.$$

Changing values at isolated points does not change an L^2 vector.

Worked example 3.1.2 (A square-integrable singularity)**.**
On $I = (0,1)$, the function $f(x) = x^{-1/4}$ lies in L^2 because

$$\int_0^1 |f(x)|^2 \, \mathrm{d}x = \int_0^1 x^{-1/2} \, \mathrm{d}x = 2.$$

The function $x^{-1/2}$ does not lie in $L^2(0,1)$ because the resulting inte-

gral $\int_0^1 x^{-1}\,\mathrm{d}x$ diverges.

3.2. Linear combinations, span and linear independence

Functions $f_1, \dots, f_k$ are linearly independent when $\sum_{j=1}^{k} \alpha_j f_j(x) = 0$ for almost every x implies all $\alpha_j = 0$.

Worked example 3.2.1 (Independent functions)**.**
The functions $1, x, x^2$ are linearly independent in $L^2(0,1)$. If $a + bx + cx^2 = 0$ almost everywhere, continuity makes the polynomial zero everywhere, so $a = b = c = 0$.

Finite spans produce polynomial, finite-element and Fourier approximations.

3.3. Bases and coordinates

An orthonormal basis of $L^2(I)$ is a sequence of functions $(\phi_n)_{n\ge 1}$ satisfying

$$\int_I \phi_n(x)\phi_m(x)\,\mathrm{d}x = \delta_{nm},$$

whose convergent linear combinations approximate every L^2 function. The coordinate label x is continuous; the basis may still be countable.

Worked example 3.3.1 (Sine basis)**.**
On $(0, \pi)$,

$$\phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx), \qquad n = 1, 2, \ldots,$$

is an orthonormal basis of $L^2(0, \pi)$. The coefficient of f in direction ϕ_n is

$$c_n = \langle \phi_n, f \rangle = \int_0^\pi \phi_n(x) f(x)\, \mathrm{d}x.$$

3.4. Decomposition and convergence

For an orthonormal basis (ϕ_n),

$$\boxed{f = \sum_{n=1}^{\infty} c_n \phi_n, \qquad c_n = \langle \phi_n, f \rangle},$$

where equality means convergence in L^2 norm:

$$\left\| f - \sum_{n=1}^{N} c_n \phi_n \right\|_2 \longrightarrow 0.$$

L^2 convergence need not be pointwise convergence.

Worked example 3.4.1 (Fourier sine expansion of a constant)**.** Let $f(x) = 1$ on $(0, \pi)$. In the basis above,

$$c_n = \sqrt{\frac{2}{\pi}} \int_0^\pi \sin(nx)\,\mathrm{d}x = \sqrt{\frac{2}{\pi}} \frac{1 - (-1)^n}{n}.$$

Thus only odd modes occur, and in $L^2(0, \pi)$,

$$1 = \frac{4}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \cdots\right).$$

At the interval endpoints the sine series takes the boundary value zero; this does not contradict L^2 equality because individual points have zero measure.

3.5. Linear operators

A linear operator satisfies $A(\alpha f + \beta g) = \alpha A f + \beta A g$. Examples are

$$(M_V f)(x) = V(x) f(x),$$

$$(D f)(x) = f'(x),$$

$$(K f)(x) = \int_I K(x, y) f(y)\,\mathrm{d}y.$$

M_V is bounded when V is essentially bounded. Differentiation is unbounded and requires a domain. Kernel conditions control K.

Worked example 3.5.1 (A bounded multiplication operator)**.**
On $L^2(0,1)$ let $(Af)(x) = xf(x)$. Since $|x| \le 1$,

$$\|Af\|_2^2 = \int_0^1 x^2 |f(x)|^2 \, \mathrm{d}x \le \|f\|_2^2 .$$

Hence A is bounded and $\|A\| = 1$.

3.6. Kernels and operator action

The matrix rule can be read one index set at a time:

$$\boxed{\begin{aligned} (Av)_i = \sum_j A_{ij} v_j \quad &\longrightarrow \quad (Aa)_n = \sum_m A_{nm} a_m, \\ (Af)_x = \sum_y (A_{xy} f_y) \quad &\rightsquigarrow \quad \int_I A(x,y) f(y) \, \mathrm{d}y. \end{aligned}}$$

A kernel $A(x,y)$ is a continuously indexed matrix. The symbol $\sum_y$ is again heuristic.

Worked example 3.6.1 (A rank-one kernel)**.**
On $L^2(0,1)$ let $A(x,y) = xy$. Then

$$(Af)(x) = \int_0^1 xy f(y) \, \mathrm{d}y = x \int_0^1 y f(y) \, \mathrm{d}y.$$

Every output is a multiple of the single function x, so the range is one-dimensional. For $f(y) = 1$, $(Af)(x) = x/2$; for $f(y) = y$, $(Af)(x) =$

$x/3$.

Worked example 3.6.2 (Composition resembles matrix multiplication)**.**
If A and B have kernels $A(x, y)$ and $B(y, z)$, then formally

$$(AB)(x, z) = \int_I A(x, y)B(y, z)\,\mathrm{d}y,$$

the exact analogue of $(AB)_{ik} = \sum_j A_{ij}B_{jk}$.

3.7. Inner products

With the continuously indexed vector picture in view, write first

$$\langle f, g\rangle = \sum_x (f_x g_x) \quad \rightsquigarrow \quad \int_I f_x g_x\,\mathrm{d}x.$$

The rigorous definition is therefore:

Definition 3.7.1.
For real $f, g \in L^2(I)$,

$$\boxed{\langle f, g\rangle = \int_I f(x)g(x)\,\mathrm{d}x}.$$

The integral is finite by the integral form of Cauchy–Schwarz.

Worked example 3.7.2 (An explicit inner product)**.**
On $(0,1)$, let $f(x) = x$ and $g(x) = 1 - x$. Then

$$\langle f, g\rangle = \int_0^1 x(1-x)\,\mathrm{d}x = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

The functions are not orthogonal, but the calculation quantifies their overlap.

3.8. Norms and p-norms

The heuristic continuous-coordinate formula is

$$\|f\|_2^2 = \sum_x (f_x^2) \quad \rightsquigarrow \quad \int_I f_x^2\,\mathrm{d}x.$$

In standard notation,

$$\boxed{\|f\|_p = \left(\int_I |f(x)|^p\,\mathrm{d}x\right)^{1/p}}, \quad 1 \le p < \infty, \qquad \|f\|_\infty = \operatorname*{ess\,sup}_{x\in I} |f(x)|.$$

Worked example 3.8.1 (Three norms of one function)**.**
For $f(x) = x$ on $(0,1)$,

$$\|f\|_1 = \frac{1}{2}, \qquad \|f\|_2 = \frac{1}{\sqrt{3}}, \qquad \|f\|_\infty = 1.$$

These norms are not generally equivalent.

3.9. Normalisation and orthogonality

For $f \neq 0$, set $\widehat{f} = f/\|f\|_2$.

Functions are orthogonal when $\int_I f(x)g(x)\,\mathrm{d}x = 0$.

Worked example 3.9.1 (Normalising a constant state)**.**
On an interval $(0, L)$, the constant function 1 has norm $\sqrt{L}$. Its normalised form is

$$\phi_0(x) = \frac{1}{\sqrt{L}}.$$

This is the normalised constant mode.

Worked example 3.9.2 (Orthogonality of sine modes)**.**
On $(0, \pi)$,

$$\int_0^{\pi} \sin(nx)\sin(mx)\,\mathrm{d}x = 0 \qquad (n \neq m).$$

After multiplication by $\sqrt{2/\pi}$, the modes are normalised as well as orthogonal.

3.10. Orthonormal bases and projections

For a normalised function ϕ, projection onto its span is

$$P_\phi f = \langle \phi, f \rangle \, \phi,$$

$$(P_\phi f)(x) = \phi(x) \int_I \phi(y) f(y) \, \mathrm{d}y.$$

The projection kernel is $P_\phi(x, y) = \phi(x)\phi(y)$. For an orthonormal family, projection onto its first N members is

$$P_N f = \sum_{n=1}^{N} \langle \phi_n, f \rangle \, \phi_n.$$

Worked example 3.10.1 (Best one-mode approximation)**.**
Project $f(x) = x$ on $(0, 1)$ onto the normalised constant $\phi_0(x) = 1$. Then

$$P_0 f = \langle 1, x \rangle \; 1 = \frac{1}{2}.$$

Among all constants c, the value $c = 1/2$ uniquely minimises $\int_0^1 |x - c|^2 \, \mathrm{d}x$.

Motivational bridge

Projection is the same operation in every part:

$$P_i v = \langle e_i, v\rangle\, e_i, \qquad P_n a = \langle e_n, a\rangle\, e_n, \qquad P_\phi f = \langle \phi, f\rangle\, \phi.$$

Only the inner product formula changes.

3.11. Failure of equivalence of norms in infinite dimensions

Theorem 3.11.1 (Failure of L^p norm equivalence)**.**
On infinite-dimensional function spaces, different L^p norms are not generally equivalent. Even on the finite interval $(0,1)$ there are no constants $c, C > 0$ such that

$$c\,\|f\|_1 \le \|f\|_2 \le C\,\|f\|_1$$

for all functions in a common dense class.

Proof. Consider the increasingly concentrated spike

$$f_n(x) = \sqrt{n}\,\mathbf{1}_{(0,1/n)}(x).$$

Then

$$\|f_n\|_2^2 = \int_0^{1/n} n\,\mathrm{d}x = 1, \qquad \|f_n\|_1 = \int_0^{1/n} \sqrt{n}\,\mathrm{d}x = \frac{1}{\sqrt{n}}.$$

If $\|f\|_2 \le C\,\|f\|_1$ held uniformly, we would have $1 \le C/\sqrt{n}$ for every n, which is impossible. □

Worked example 3.11.2 (The underlying spaces may differ)**.**
On $(0,1)$, $f(x) = x^{-1/2}$ belongs to L^1 but not to L^2:

$$\int_0^1 x^{-1/2}\,\mathrm{d}x = 2, \qquad \int_0^1 x^{-1}\,\mathrm{d}x = \infty.$$

Thus, as with sequence spaces, changing p may change the admissible vectors, not merely their numerical lengths.

Important distinction

On finite-measure intervals, $L^2(I) \subset L^1(I)$. This one-way estimate is not norm equivalence.

3.12. Resolution of the identity and Parseval's identity

Proposition 3.12.1 (Basis expansion)**.**
For a complete orthonormal basis (ϕ_n)*,*

$$f = \sum_{n=1}^{\infty} \langle \phi_n, f\rangle\, \phi_n$$

in the L^2 norm. Equivalently,

$$\boxed{\mathrm{I} = \sum_{n=1}^{\infty} P_n}.$$

Proposition 3.12.2 (Parseval's identity)**.**
For every $f \in L^2(I)$,

$$\boxed{\|f\|_2^2 = \sum_{n=1}^{\infty} |\,\langle \phi_n, f\rangle\,|^2}.$$

In continuous-coordinate language, the identity operator has the formal kernel $\delta(x-y)$:

$$f(x) = \int_I \delta(x-y)\, f(y)\,\mathrm{d}y.$$

The Dirac delta is a generalised function, not an element of L^2.

Worked example 3.12.3 (Parseval evaluates a series)**.**
For $f(x) = 1$ on $(0,\pi)$ and the sine basis of Section 3.3, $\|f\|_2^2 = \pi$ and

$$|c_n|^2 = \begin{cases} 8/(\pi n^2), & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Parseval therefore gives

$$\pi = \frac{8}{\pi}\sum_{n\ \text{odd}}\frac{1}{n^2}, \qquad \sum_{n\ \text{odd}}\frac{1}{n^2} = \frac{\pi^2}{8}.$$

3.13. Eigenvalues and eigenfunctions

An eigenfunction is a non-zero function in the operator domain satisfying

$$\boxed{Af = \lambda f}.$$

This is the usual eigenvector equation with a function as the vector.

Worked example 3.13.1 (A boundary-value eigenproblem)**.**
On $(0,\pi)$ consider

$$Af = -f'', \qquad f(0) = f(\pi) = 0.$$

The functions $f_n(x) = \sin(nx)$ satisfy

$$Af_n = n^2 f_n.$$

Thus $\lambda_n = n^2$ and $\phi_n(x) = \sqrt{2/\pi}\sin(nx)$. Boundary conditions change the spectrum.

3.14. Eigenspaces and degeneracy

The eigenspace is again $E_\lambda = \ker(A - \lambda \mathrm{I})$. Degeneracy means $\dim E_\lambda > 1$.

Worked example 3.14.1 (Degeneracy on a periodic interval)**.**
On real periodic functions over $(-\pi, \pi)$, let $A = -\mathrm{d}^2/\mathrm{d}x^2$ with periodic boundary conditions. For each $n \geq 1$,

$$A\cos(nx) = n^2\cos(nx), \qquad A\sin(nx) = n^2\sin(nx).$$

Thus n^2 has degeneracy two. The eigenspace for 0 consists of constant functions and has dimension one.

3.15. Adjoints and self-adjoint operators

The adjoint is still defined by

$$\boxed{\langle f, Ag\rangle = \langle A^* f, g\rangle}.$$

For a kernel operator,

$$(Af)(x) = \int_I A(x,y) f(y)\,\mathrm{d}y \quad \Longrightarrow \quad (A^* f)(x) = \int_I A(y,x) f(y)\,\mathrm{d}y$$

under suitable hypotheses. A symmetric real kernel $A(x,y) = A(y,x)$ therefore suggests a self-adjoint operator.

For differential operators, integrate by parts:

$$\int_0^\pi f(-g'')\,\mathrm{d}x = \int_0^\pi (-f'')g\,\mathrm{d}x + \left[f'g - fg'\right]_0^\pi.$$

Dirichlet conditions $f(0) = f(\pi) = g(0) = g(\pi) = 0$ remove the boundary term.

Important distinction

The differential expression, domain and boundary conditions define the operator together. Integration by parts tests formal symmetry; full self-adjointness also requires a domain argument.

Worked example 3.15.1 (A symmetric kernel)**.**
The kernel $A(x, y) = \min(x, y)$ on $(0, 1)$ satisfies $A(x, y) = A(y, x)$, so the associated integral operator is self-adjoint. By contrast, a non-symmetric kernel and its transpose generally define different adjoint operators.

3.16. Real eigenvalues of self-adjoint operators

Proposition 3.16.1.
If A is self-adjoint and $Af = \lambda f$ for non-zero $f \in \mathcal{D}(A)$, then

$$\boxed{\lambda = \frac{\langle f, Af\rangle}{\langle f, f\rangle} \in \mathbb{R}}.$$

Proof. We have $\langle f, Af\rangle = \lambda \langle f, f\rangle$. The numerator is real and the denominator is positive. □

Worked example 3.16.2 (Rayleigh quotient for $-f''$)**.**
For a sufficiently smooth real function with $f(0) = f(\pi) = 0$, integration by parts gives

$$\frac{\langle f, -f''\rangle}{\langle f, f\rangle} = \frac{\int_0^\pi |f'(x)|^2 \,\mathrm{d}x}{\int_0^\pi |f(x)|^2 \,\mathrm{d}x} \geq 0.$$

Thus the Dirichlet eigenvalues of $-\mathrm{d}^2/\mathrm{d}x^2$ are not only real but non-negative.

3.17. Orthogonality of eigenfunctions

Theorem 3.17.1.
Eigenfunctions of a self-adjoint operator belonging to distinct eigen-

values are orthogonal in L^2.

Proof. If $Af = \lambda f$ and $Ag = \mu g$, then

$$\mu \langle f, g \rangle = \langle f, Ag \rangle = \langle Af, g \rangle = \lambda \langle f, g \rangle .$$

For $\lambda \neq \mu$, this gives $\langle f, g \rangle = 0$. □

Worked example 3.17.2 (Sturm–Liouville orthogonality)**.**
For the Dirichlet eigenfunctions $\sin(nx)$ and $\sin(mx)$ on $(0, \pi)$, distinct eigenvalues $n^2 \neq m^2$ imply

$$\int_0^\pi \sin(nx) \sin(mx)\, \mathrm{d}x = 0.$$

The familiar trigonometric identity is therefore an instance of a general self-adjoint principle.

3.18. Spectral theorem and spectral issues

Compact self-adjoint kernels have orthonormal eigenfunction bases. Other self-adjoint operators may have continuous spectrum.

Worked example 3.18.1 (Continuous spectrum without L^2 eigenfunctions)**.**

On $L^2(-1,1)$ define $(Mf)(x) = xf(x)$. This bounded operator is self-adjoint. If $Mf = \lambda f$, then

$$(x - \lambda) f(x) = 0$$

almost everywhere. Hence f could be non-zero only at the single point $x = \lambda$, which represents the zero vector in L^2. Thus M has no non-zero L^2 eigenfunctions, even though its spectrum is the entire interval $[-1,1]$.

Thus self-adjointness need not give an eigenbasis. Spectral projections replace a countable list of eigenvectors.

Important distinction

Generalised eigenfunctions need not lie in L^2. Their proper setting is distribution and spectral theory.

3.19. Spectral decomposition

In a discrete eigenfunction basis,

$$\boxed{Af = \sum_{n=1}^{\infty} \lambda_n \langle \phi_n, f \rangle \phi_n}, \qquad A = \sum_{n=1}^{\infty} \lambda_n P_n.$$

In general,

$$\boxed{A = \int_{\sigma(A)} \lambda \, \mathrm{d}P(\lambda)}.$$

Worked example 3.19.1 (Dirichlet Laplacian)**.**
Let $A = -\mathrm{d}^2/\mathrm{d}x^2$ on $(0, \pi)$. Impose Dirichlet boundary conditions. Then

$$Af = \sum_{n=1}^{\infty} n^2 \, \langle \phi_n, f \rangle \, \phi_n, \qquad \phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx),$$

for functions in the appropriate operator domain. Solving $u_t = -Au$ mode by mode gives

$$u(t) = \sum_{n=1}^{\infty} \mathrm{e}^{-n^2 t} \, \langle \phi_n, u(0) \rangle \, \phi_n,$$

which is the spectral solution of the heat equation.

Worked example 3.19.2 (Continuous decomposition of multiplication)**.**
For $Mf(x) = xf(x)$ on $L^2(-1, 1)$, define a projection for each measurable set $B \subset [-1, 1]$ by

$$(P(B)f)(x) = \mathbf{1}_B(x) f(x).$$

Then $M = \int_{-1}^{1} \lambda \, \mathrm{d}P(\lambda)$. No ordinary eigenbasis is required.

$$f_x \longrightarrow f(x), \qquad \sum_x (f_x g_x) \rightsquigarrow$$

$$\int f(x)g(x)\,\mathrm{d}x, \qquad (Af)_x = \sum_y (A_{xy} f_y) \rightsquigarrow$$

$$\int A(x,y)f(y)\,\mathrm{d}y.$$

Part IV.

Fourier Coordinates and Evolution Equations

4. Fourier Coordinates, Heat Flow and Quantum Motion

The guiding idea is the finite-dimensional rule

choose a useful basis $\quad\Longrightarrow\quad$ study its coordinates.

If the basis consists of eigenvectors, an operator acts by multiplying coordinates. This part develops that idea slowly for functions.

4.1. Moving vectors and moving functions

Begin with an ordinary moving vector:

$$v(t) = \cos(t)\,\mathbf{i} + \sin(t)\,\mathbf{j}.$$

The basis vectors $\mathbf{i}$ and $\mathbf{j}$ are fixed.

The coordinates move:

$$v_1(t) = \cos t, \qquad v_2(t) = \sin t.$$

At each time t, the pair

$$\begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

describes one vector. As t changes, the vector traces a circle.

Motivational bridge

Time dependence does not require a new kind of vector.
It means that the coefficients in a fixed basis now depend on time:

$$v(t) = \sum_i c_i(t) e_i.$$

The same interpretation applies to a function $u(x,t)$.

For each fixed t, the map

$$x \longmapsto u(x,t)$$

is one vector in a function space.

Heuristically write

$$u(t) = \big(u_x(t)\big)_{x\in\mathbb{R}}, \qquad u_x(t) = u(x,t).$$

The number $u(x,t)$ is the coefficient in the direction labelled by x.

Time changes all these coefficients:

$$t \longmapsto u(t).$$

Definition 4.1.1 (A time-dependent function-vector)**.**
A function $u(x,t)$ may be viewed as a path through a function space.

For each fixed t:

$$u(\,\cdot\,,t)$$

is one vector.

For each fixed x:

$$u(x,t)$$

is one time-dependent coefficient.

If a fixed orthonormal basis (ϕ_n) is used, then

$$u(x,t) = \sum_n c_n(t)\phi_n(x).$$

The basis functions do not move. The coordinates $c_n(t)$ do.

Consequently,

$$\frac{\partial u}{\partial t}(x,t) = \sum_n c_n'(t)\phi_n(x).$$

This simple observation is the heart of the later heat and Schrödinger calculations.

4.2. Position directions and narrow boxes

For $v \in \mathbb{R}^N$, the standard basis vector e_i represents the ith coordinate direction:

$$v_i = \langle e_i, v \rangle , \qquad v = \sum_i v_i e_i .$$

For a function, imagine one coordinate direction for every point x:

$$f = (f_x)_{x \in \mathbb{R}}, \qquad f_x = f(x).$$

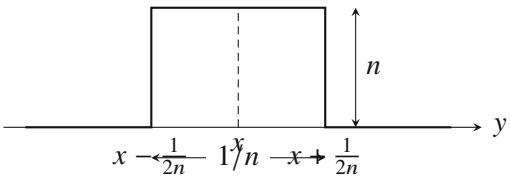
A narrow box gives a visible approximation to such a direction.

Definition 4.2.1 (A unit-area box centred at x)**.**
For $n = 1, 2, \ldots$, define

$$\psi_{x,n}(y) = \begin{cases} n, & x - \dfrac{1}{2n} < y < x + \dfrac{1}{2n}, \\ 0, & \text{otherwise.} \end{cases}$$

Its width is $1/n$, its height is n, and its area is one:

$$\int_{-\infty}^{\infty} \psi_{x,n}(y)\, \mathrm{d}y = 1.$$



A narrower box is also taller, so its area remains one.

If f varies little across the box, then

$$\int_{-\infty}^{\infty} \psi_{x,n}(y) f(y)\,\mathrm{d}y \approx f(x).$$

The box approximately extracts the coefficient at x.

Worked example 4.2.2 (Extracting a local value)**.**
Let $f(y) = y^2$ and let the box have width h.

Its average is

$$\frac{1}{h}\int_{x-h/2}^{x+h/2} y^2\,\mathrm{d}y = x^2 + \frac{h^2}{12}.$$

As $h \to 0$, this tends to $x^2 = f(x)$.

4.3. The delta direction

The idealised coordinate direction at x is written δ_x.

Definition 4.3.1 (Delta direction at x: motivational form)**.**
The symbol $\delta_x(y)$ is characterised heuristically by

$$\boxed{\int_{-\infty}^{\infty} \delta_x(y) f(y)\,\mathrm{d}y = f(x).}$$

It plays the role of the basis vector for the direction labelled by x.

The narrow boxes motivate the notation

$$\psi_{x,n}(y) \longrightarrow \delta_x(y).$$

No precise meaning of this limit is required here.

Just as e_i extracts v_i, the delta direction extracts f_x:

$$\langle e_i, v \rangle = v_i, \qquad \langle \delta_x, f \rangle = f(x).$$

The corresponding expansions are

$$v = \sum_i v_i e_i,$$

and, heuristically,

$$\boxed{f(y) = \int_{-\infty}^{\infty} f(x)\delta_x(y)\,\mathrm{d}x.}$$

Moving the direction centred at zero

First define the direction at the origin:

$$\delta_0(y) = \delta(y).$$

Translate it from 0 to x:

$$\boxed{\delta_x(y) = \delta_0(y - x).}$$

The usual delta notation therefore gives

$$\delta_x(y) = \delta(y - x).$$

Since the delta is symmetric,

$$\boxed{\delta_x(y) = \delta(y - x) = \delta(x - y).}$$

This explains why an identity kernel is written

$$I(x, y) = \delta(x - y).$$

Indeed,

$$(If)(x) = \int \delta(x - y) f(y)\, \mathrm{d}y = f(x).$$

Important distinction

The delta direction is not an ordinary L^2 function.
The unit-area boxes also satisfy

$$\|\psi_{x,n}\|_2 = \sqrt{n}.$$

Their norms grow as they become narrower. The delta notation is therefore a coordinate picture, not an orthonormal basis inside L^2.

A genuine orthonormal box approximation is obtained by dividing the line into disjoint cells I_j of width h and setting

$$\phi_{j,h}(y) = \frac{1}{\sqrt{h}} \mathbf{1}_{I_j}(y).$$

Then

$$\langle \phi_{j,h}, \phi_{k,h} \rangle = \delta_{jk}.$$

The coordinate is

$$c_j = \langle \phi_{j,h}, f \rangle = \frac{1}{\sqrt{h}} \int_{I_j} f(y)\, \mathrm{d}y.$$

The reconstructed term is

$$c_j \phi_{j,h}(y) = \left(\frac{1}{h} \int_{I_j} f(s)\, \mathrm{d}s \right) \mathbf{1}_{I_j}(y).$$

Thus each cell is replaced by its average value.

4.4. Why choose a Fourier basis?

The position directions describe where a function is located.

For differentiation, oscillating directions are more useful:

$$e^{inx}.$$

Differentiation does not change their shape:

$$\frac{d}{dx}e^{inx} = in\, e^{inx}.$$

This is the function-space analogue of choosing eigenvectors of a matrix.

In an eigenvector basis,

$$A\left(\sum_i c_i q_i\right) = \sum_i \lambda_i c_i q_i.$$

In a Fourier basis, a differential operator will likewise multiply each coefficient.

4.5. The Fourier basis on $L^2(-\pi, \pi)$

Complex notation combines sine and cosine into one expression.

Use the inner product

$$\langle g, f \rangle = \int_{-\pi}^{\pi} \overline{g(x)} f(x)\, \mathrm{d}x.$$

Definition 4.5.1 (Normalised complex Fourier functions)**.**
For $n \in \mathbb{Z}$, define

$$\boxed{\phi_n(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}nx}.}$$

The factor $1/\sqrt{2\pi}$ makes every basis function have norm one:

$$\|\phi_n\|_2^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} 1\, \mathrm{d}x = 1.$$

Orthogonality

For $m, n \in \mathbb{Z}$,

$$\langle \phi_m, \phi_n \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{i}(n-m)x}\, \mathrm{d}x.$$

If $m = n$, the integrand is 1, so

$$\langle \phi_n, \phi_n \rangle = 1.$$

If $m \neq n$, then

$$\langle \phi_m, \phi_n \rangle = \frac{1}{2\pi} \left[\frac{\mathrm{e}^{\mathrm{i}(n-m)x}}{\mathrm{i}(n-m)} \right]_{-\pi}^{\pi}$$

$$= 0.$$

Hence

$$\boxed{\langle \phi_m, \phi_n \rangle = \delta_{mn}.}$$

Proposition 4.5.2 (Fourier coordinates)**.**
The coordinate of f in the direction ϕ_n is

$$\boxed{c_n = \langle \phi_n, f \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) \mathrm{e}^{-\mathrm{i}nx} \, \mathrm{d}x.}$$

The coordinate expansion is

$$\boxed{f(x) = \sum_{n \in \mathbb{Z}} c_n \phi_n(x)}$$

in the L^2 sense.

Motivational bridge

A Fourier coefficient is not a new kind of object.
It is simply a coordinate in the Fourier basis:

Fourier coefficient = coordinate in the Fourier basis.

The phrase *Fourier series* means the corresponding basis expansion.

A second common convention uses

$$\widehat{f_n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \mathrm{e}^{-\mathrm{i}nx} \, \mathrm{d}x.$$

Then

$$f(x) = \sum_{n \in \mathbb{Z}} \widehat{f_n} \mathrm{e}^{\mathrm{i}nx}.$$

The two conventions differ only by normalisation:

$$c_n = \sqrt{2\pi} \, \widehat{f_n}.$$

The real sine and cosine basis

Euler's formula gives

$$\mathrm{e}^{\mathrm{i}nx} = \cos(nx) + \mathrm{i} \sin(nx).$$

Also,

$$\cos(nx) = \frac{e^{inx} + e^{-inx}}{2},$$

$$\sin(nx) = \frac{e^{inx} - e^{-inx}}{2i}.$$

Thus the real and complex descriptions contain the same oscillations.

A real orthonormal basis is

$$\boxed{\frac{1}{\sqrt{2\pi}}, \qquad \frac{\cos(nx)}{\sqrt{\pi}}, \qquad \frac{\sin(nx)}{\sqrt{\pi}}, \quad n = 1, 2, \ldots.}$$

4.6. Changing the interval to $(-L, L)$

The interval now has length $2L$.

To fit exactly across this interval, the frequencies become

$$k_n = \frac{n\pi}{L}.$$

Definition 4.6.1 (Fourier basis on $(-L, L)$)**.**
For $n \in \mathbb{Z}$, define

$$\boxed{\phi_n^{(L)}(x) = \frac{1}{\sqrt{2L}} e^{in\pi x/L}.}$$

The normalisation again gives unit norm:

$$\left\|\phi_n^{(L)}\right\|_2^2 = \frac{1}{2L}\int_{-L}^{L} 1\,\mathrm{d}x = 1.$$

Orthogonality follows from

$$\left\langle \phi_m^{(L)}, \phi_n^{(L)} \right\rangle = \frac{1}{2L}\int_{-L}^{L} \mathrm{e}^{\mathrm{i}(n-m)\pi x/L}\,\mathrm{d}x = \delta_{mn}.$$

The real orthonormal basis is

$$\frac{1}{\sqrt{2L}}, \qquad \frac{\cos(n\pi x/L)}{\sqrt{L}}, \qquad \frac{\sin(n\pi x/L)}{\sqrt{L}}, \qquad n = 1, 2, \ldots.$$

Motivational bridge

Nothing essential has changed.

On $(-\pi, \pi)$:

n is the wave number.

On $(-L, L)$:

$\dfrac{n\pi}{L}$ is the wave number.

The interval length changes the allowed frequencies and the normalising constant.

4.7. Real Fourier series and three examples

For a real function on $(-L, L)$, the conventional form is

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right],$$

where

$$a_n = \frac{1}{L} \int_{-L}^{L} f(x) \cos\left(\frac{n\pi x}{L}\right) \, \mathrm{d}x,$$

$$b_n = \frac{1}{L} \int_{-L}^{L} f(x) \sin\left(\frac{n\pi x}{L}\right) \, \mathrm{d}x.$$

These a_n and b_n are again coordinates, with a conventional normalisation.

Even functions have only cosine terms.

Odd functions have only sine terms.

What happens at a jump?

Suppose f has one-sided limits at x:

$$f(x^-), \qquad f(x^+).$$

For a piecewise smooth function, its Fourier series converges there to

$$\boxed{\frac{f(x^-) + f(x^+)}{2}}.$$

At a continuous point this is simply $f(x)$.

At a jump it is the midpoint between the two limiting values.

This does not conflict with L^2 equality: changing a function at one point does not change its L^2 vector.

Worked example 4.7.1 (The function x^2 on $(-\pi, \pi)$)**.**
The function is even, so $b_n = 0$.

Its coefficients are

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\,\mathrm{d}x = \frac{2\pi^2}{3},$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x^2\cos(nx)\,\mathrm{d}x = \frac{4(-1)^n}{n^2}.$$

Therefore

$$\boxed{x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx).}$$

At $x = \pi$, this gives

$$\pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

Hence

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}.$$

Worked example 4.7.2 (A step function)**.**

On $(-\pi, \pi)$, define

$$s(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi. \end{cases}$$

The function is odd, so only sine coordinates occur:

$$\boxed{s(x) \sim \frac{4}{\pi} \sum_{r=0}^{\infty} \frac{\sin((2r+1)x)}{2r+1}.}$$

At $x = 0$,

$$s(0^-) = -1, \qquad s(0^+) = 1.$$

The Fourier series therefore takes the midpoint value

$$\frac{-1+1}{2} = 0.$$

Every sine term is also zero at $x = 0$.

At $x = \pi$, the periodic extension jumps from 1 back to -1. The series again converges to 0.

Worked example 4.7.3 (The tent function $1 - |x|$ on $(-1, 1)$)**.**
Let

$$f(x) = 1 - |x|.$$

This function is even and has period 2 after periodic extension.

Its Fourier series is

$$\boxed{1 - |x| = \frac{1}{2} + \frac{4}{\pi^2} \sum_{r=0}^{\infty} \frac{\cos((2r+1)\pi x)}{(2r+1)^2}.}$$

Only odd cosine modes occur.

At $x = 0$,

$$1 = \frac{1}{2} + \frac{4}{\pi^2} \sum_{r=0}^{\infty} \frac{1}{(2r+1)^2}.$$

At $x = 1$,

$$0 = \frac{1}{2} - \frac{4}{\pi^2} \sum_{r=0}^{\infty} \frac{1}{(2r+1)^2}.$$

Both give

$$\sum_{r=0}^{\infty} \frac{1}{(2r+1)^2} = \frac{\pi^2}{8}.$$

4.8. Parseval's identity

For an orthonormal basis in $\mathbb{R}^N$,

$$\|v\|^2 = \sum_i |c_i|^2.$$

The length of the vector equals the length of its coordinate list.

The same statement holds for Fourier coordinates.

Theorem 4.8.1 (Parseval on $(-\pi, \pi)$)**.**
If

$$f = \sum_{n \in \mathbb{Z}} c_n \phi_n,$$

then

$$\boxed{\int_{-\pi}^{\pi} |f(x)|^2 \, \mathrm{d}x = \sum_{n \in \mathbb{Z}} |c_n|^2.}$$

For the conventional real coefficients on $(-L, L)$,

$$\boxed{\frac{1}{L} \int_{-L}^{L} |f(x)|^2 \, \mathrm{d}x = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).}$$

Motivational bridge

Parseval says that changing to Fourier coordinates does not change length.
It is the function-space version of

$$P^{-1} = P^T$$

for an orthogonal change of basis in $\mathbb{R}^N$.

Worked example 4.8.2 (Parseval for the step function)**.**
For the step function, $|s(x)|^2 = 1$ almost everywhere. Hence

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |s(x)|^2 \, \mathrm{d}x = 2.$$

Its non-zero sine coefficients are

$$b_{2r+1} = \frac{4}{\pi(2r+1)}.$$

Parseval gives

$$2 = \frac{16}{\pi^2} \sum_{r=0}^{\infty} \frac{1}{(2r+1)^2},$$

again yielding

$$\sum_{r=0}^{\infty} \frac{1}{(2r+1)^2} = \frac{\pi^2}{8}.$$

4.9. Differential operators multiply Fourier coordinates

On $(-L, L)$,

$$\phi_n^{(L)}(x) = \frac{1}{\sqrt{2L}} \mathrm{e}^{\mathrm{i}n\pi x/L}.$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x}\phi_n^{(L)} = \frac{\mathrm{i}n\pi}{L}\phi_n^{(L)},$$

and

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi_n^{(L)} = \left(\frac{n\pi}{L}\right)^2 \phi_n^{(L)}.$$

If

$$f = \sum_{n\in\mathbb{Z}} c_n \phi_n^{(L)},$$

then, formally,

$$\boxed{f' = \sum_{n\in\mathbb{Z}} \left(\frac{\mathrm{i}n\pi}{L}\right) c_n \phi_n^{(L)},}$$

and

$$\boxed{-f'' = \sum_{n\in\mathbb{Z}} \left(\frac{n\pi}{L}\right)^2 c_n \phi_n^{(L)}.}$$

Thus differentiation acts directly on the coordinates:

$$c_n \longmapsto \frac{\mathrm{i}n\pi}{L} c_n,$$

$$c_n \longmapsto \left(\frac{n\pi}{L}\right)^2 c_n.$$

Motivational bridge

Compare the three calculations:

$$Aq_i = \lambda_i q_i, \qquad c_i \longmapsto \lambda_i c_i,$$

$$D\phi_n^{(L)} = \frac{\mathrm{i}n\pi}{L}\phi_n^{(L)}, \quad c_n \longmapsto \frac{\mathrm{i}n\pi}{L} c_n.$$

The operator is complicated in the original coordinates but simple in its eigenbasis.

Worked example 4.9.1 (Differentiate one Fourier mode)**.**
On $(-\pi, \pi)$,

$$f(x) = \cos(3x) = \frac{1}{2}\mathrm{e}^{3\mathrm{i}x} + \frac{1}{2}\mathrm{e}^{-3\mathrm{i}x}.$$

Differentiation multiplies the two coordinates by 3i and -3i:

$$f'(x) = \frac{3\mathrm{i}}{2}\mathrm{e}^{3\mathrm{i}x} - \frac{3\mathrm{i}}{2}\mathrm{e}^{-3\mathrm{i}x}.$$

Therefore

$$f'(x) = -3\sin(3x).$$

4.10. The periodic heat equation

Consider the heat equation on $(-L, L)$:

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}, \qquad u(x,0) = f(x),$$

with periodic boundary conditions.

At each time t, expand the function-vector:

$$\boxed{u(x,t) = \sum_{n\in\mathbb{Z}} c_n(t)\phi_n^{(L)}(x).}$$

This is exactly the form

$$v(t) = \sum_i c_i(t)e_i$$

from finite-dimensional linear algebra.

The coefficients are now time-dependent.

Differentiate with respect to time:

$$u_t = \sum_{n\in\mathbb{Z}} c_n'(t)\phi_n^{(L)}.$$

Differentiate twice with respect to position:

$$u_{xx} = -\sum_{n\in\mathbb{Z}} \left(\frac{n\pi}{L}\right)^2 c_n(t)\phi_n^{(L)}.$$

Equality of the two function-vectors means equality of every coordinate:

$$c_n'(t) = -\kappa \left(\frac{n\pi}{L}\right)^2 c_n(t).$$

The partial differential equation has become one scalar ordinary differential equation for each n.

Theorem 4.10.1 (Periodic heat flow)**.**
Let

$$c_n(0) = \left\langle \phi_n^{(L)}, f \right\rangle.$$

Then

$$\boxed{c_n(t) = \exp\left[-\kappa \left(\frac{n\pi}{L}\right)^2 t\right] c_n(0).}$$

Hence

$$\boxed{u(x,t) = \sum_{n\in\mathbb{Z}} \exp\left[-\kappa \left(\frac{n\pi}{L}\right)^2 t\right] c_n(0)\phi_n^{(L)}(x).}$$

The $n = 0$ coordinate is unchanged.

Coordinates with large $|n|$ decay quickly.

Large $|n|$ corresponds to rapid oscillation, so heat flow removes fine detail first.

Worked example 4.10.2 (Two temperature modes)**.**
On $(-\pi, \pi)$, suppose

$$u(x, 0) = 3 + 2\cos x + \frac{1}{2}\cos(4x).$$

Then

$$\boxed{u(x, t) = 3 + 2\mathrm{e}^{-\kappa t}\cos x + \frac{1}{2}\mathrm{e}^{-16\kappa t}\cos(4x).}$$

The fourth mode decays much faster than the first.

The constant coordinate 3 is the mean temperature and remains fixed.

Motivational bridge

Let

$$A = -\frac{\partial^2}{\partial x^2}.$$

Then the heat equation is

$$u_t = -\kappa A u.$$

This mirrors the finite-dimensional equation

$$v'(t) = -\kappa A v(t).$$

In an eigenbasis of A, each coordinate evolves independently:

$$c_n'(t) = -\kappa \lambda_n c_n(t).$$

4.11. From a Fourier series to a Fourier transform

Now enlarge the interval:

$$(-L, L) \longrightarrow \mathbb{R}.$$

On $(-L, L)$, the wave numbers are

$$k_n = \frac{n\pi}{L}.$$

The spacing between neighbouring wave numbers is

$$\Delta k = \frac{\pi}{L}.$$

As $L \to \infty$,

$$\Delta k \to 0.$$

The discrete frequency label n is replaced by a continuous label k.

For a function concentrated well inside $(-L, L)$,

$$c_n = \frac{1}{\sqrt{2L}} \int_{-L}^{L} f(x) \mathrm{e}^{-\mathrm{i}k_n x} \, \mathrm{d}x.$$

If

$$\widehat{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \mathrm{e}^{-\mathrm{i}kx} \, \mathrm{d}x,$$

then heuristically

$$c_n \approx \sqrt{\Delta k} \, \widehat{f}(k_n).$$

Also,

$$\phi_n^{(L)}(x) = \sqrt{\frac{\Delta k}{2\pi}}\, \mathrm{e}^{\mathrm{i}k_n x}.$$

Therefore

$$\sum_n c_n \phi_n^{(L)}(x) \approx \sum_n \frac{\Delta k}{\sqrt{2\pi}} \widehat{f}(k_n) \mathrm{e}^{\mathrm{i}k_n x}.$$

As the spacing tends to zero, the sum suggests the integral

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{f}(k) \mathrm{e}^{\mathrm{i}kx}\, \mathrm{d}k.$$

Important distinction

This passage from $L^2(-L, L)$ to $L^2(\mathbb{R})$ is motivational.
It correctly predicts the formulas, but a rigorous proof requires care about limits, convergence and the fact that plane waves are not in $L^2(\mathbb{R})$.

4.12. Fourier coordinates on $L^2(\mathbb{R})$

The position picture uses the continuous family δ_x.

The frequency picture uses the continuous family

$$\phi_k(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{\mathrm{i}kx}, \qquad k \in \mathbb{R}.$$

These plane waves are not ordinary $L^2(\mathbb{R})$ vectors. They are generalised basis functions.

Heuristically,

$$\langle \phi_k, \phi_q \rangle = \delta(k - q).$$

Definition 4.12.1 (Fourier transform)**.**
The Fourier coefficient in the direction ϕ_k is

$$\boxed{\widehat{f}(k) = \langle \phi_k, f \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \mathrm{e}^{-\mathrm{i}kx} \, \mathrm{d}x.}$$

The function $k \mapsto \widehat{f}(k)$ is called the Fourier transform.

Motivational bridge

The terminology changes, but the object does not.
For a discrete basis:

$$c_n = \langle \phi_n, f \rangle \quad \text{is a Fourier coefficient.}$$

For a continuous basis:

$$\widehat{f}(k) = \langle \phi_k, f \rangle \quad \text{is a Fourier coefficient.}$$

The Fourier transform is simply the continuous list of Fourier coefficients.

Definition 4.12.2 (Inverse Fourier transform)**.**
The continuous basis expansion is

$$\boxed{f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{f}(k) \mathrm{e}^{\mathrm{i}kx} \, \mathrm{d}k.}$$

Equivalently,

$$f = \int_{-\infty}^{\infty} \widehat{f}(k) \phi_k \, \mathrm{d}k.$$

Compare:

$$v = \sum_i c_i e_i,$$

$$f = \sum_n c_n \phi_n,$$

$$f = \int \widehat{f}(k) \phi_k \, \mathrm{d}k.$$

Only the index set has changed.

The continuous Parseval identity is

$$\boxed{\int_{-\infty}^{\infty} |f(x)|^2 \, \mathrm{d}x = \int_{-\infty}^{\infty} |\widehat{f}(k)|^2 \, \mathrm{d}k.}$$

Thus the Fourier transform preserves the length of the vector.

4.13. Differential operators in Fourier coordinates

The plane waves are eigenfunctions of differentiation:

$$\frac{\mathrm{d}}{\mathrm{d}x}\phi_k = \mathrm{i}k\phi_k,$$

$$-\frac{\mathrm{d}^2}{\mathrm{d}x^2}\phi_k = k^2\phi_k.$$

Therefore differentiation multiplies Fourier coefficients.

Proposition 4.13.1 (Fourier multipliers)**.**
When boundary terms vanish,

$$\boxed{\widehat{f'}(k) = \mathrm{i}k\,\widehat{f}(k),}$$

and

$$\boxed{\widehat{-f''}(k) = k^2\widehat{f}(k).}$$

More generally,

$$\widehat{p(D)f}(k) = p(\mathrm{i}k)\widehat{f}(k), \qquad D = \frac{\mathrm{d}}{\mathrm{d}x}.$$

The finite and continuous calculations are identical:

$$c_i \longmapsto \lambda_i c_i,$$

$$c_n \longmapsto \lambda_n c_n,$$

$$\widehat{f}(k) \longmapsto \lambda(k)\widehat{f}(k).$$

4.14. The Gaussian transforms to a Gaussian

Proposition 4.14.1 (Gaussian Fourier transform)**.**
For $a > 0$,

$$f(x) = \mathrm{e}^{-ax^2}$$

has Fourier transform

$$\boxed{\widehat{f}(k) = \frac{1}{\sqrt{2a}}\mathrm{e}^{-k^2/(4a)}.}$$

Proof. The standard Gaussian integral gives

$$\int_{-\infty}^{\infty} \mathrm{e}^{-ax^2 - \mathrm{i}kx}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}\,\mathrm{e}^{-k^2/(4a)}.$$

Multiply by $1/\sqrt{2\pi}$. □

For the unit-area Gaussian

$$g_\sigma(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\!\left(-\frac{x^2}{2\sigma^2}\right),$$

one obtains

$$\boxed{\widehat{g}_\sigma(k) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\sigma^2k^2/2}.}$$

A narrow Gaussian in position has a broad Fourier transform.

A broad Gaussian in position has a narrow Fourier transform.

4.15. The heat equation on the real line

Consider

$$u_t = \kappa u_{xx}, \qquad x \in \mathbb{R}, \qquad u(x,0) = f(x).$$

For each time t, take the Fourier coefficient in the k direction:

$$\widehat{u}(k,t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} u(x,t) \mathrm{e}^{-\mathrm{i}kx} \, \mathrm{d}x.$$

The transform of u_{xx} is

$$-k^2 \widehat{u}(k,t).$$

Hence every Fourier coordinate satisfies

$$\frac{\partial \widehat{u}}{\partial t}(k,t) = -\kappa k^2 \widehat{u}(k,t).$$

This is the continuous version of

$$c_n'(t) = -\kappa \lambda_n c_n(t).$$

Theorem 4.15.1 (Heat flow on $\mathbb{R}$)**.**

The Fourier coefficients evolve by

$$\boxed{\widehat{u}(k,t) = \mathrm{e}^{-\kappa k^2 t}\,\widehat{f}(k).}$$

After inverse transformation,

$$\boxed{u(x,t) = (G_t * f)(x),}$$

where

$$\boxed{G_t(x) = \frac{1}{\sqrt{4\pi\kappa t}}\exp\!\left(-\frac{x^2}{4\kappa t}\right).}$$

The Gaussian calculation shows that

$$\widehat{G}_t(k) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\kappa k^2 t}.$$

Thus convolution with G_t produces exactly the required multiplier.

Motivational bridge

The kernel G_t has total area one:

$$\int_{\mathbb{R}} G_t(x)\,\mathrm{d}x = 1.$$

It is a smooth averaging box.
The value $u(x,t)$ is a weighted average of the initial temperature near x. The averaging width grows like

$$\sqrt{\kappa t}.$$

If the initial temperature is the ideal position direction δ_0, then

$$u(x,t) = G_t(x).$$

A point of heat spreads into a Gaussian.

Worked example 4.15.2 (A Gaussian spreads)**.**
If

$$f(x) = g_\sigma(x),$$

then

$$\boxed{u(x,t) = g_{\sqrt{\sigma^2+2\kappa t}}(x).}$$

The total heat remains one.

The variance changes by

$$\sigma^2 \longmapsto \sigma^2 + 2\kappa t.$$

4.16. The infinite square well

Quantum wavefunctions are complex function-vectors on $(0, L)$:

$$\langle \phi, \psi \rangle = \int_0^L \overline{\phi(x)} \psi(x) \, \mathrm{d}x.$$

Definition 4.16.1 (Infinite potential well)**.**
The potential is zero for $0 < x < L$ and infinite outside.

The wavefunction obeys

$$\psi(0, t) = \psi(L, t) = 0.$$

Inside the well,

$$\boxed{H = -\frac{\hbar^2}{2m} \frac{\mathrm{d}^2}{\mathrm{d}x^2}.}$$

First find a basis adapted to H:

$$H\phi = E\phi.$$

Thus

$$-\frac{\hbar^2}{2m} \phi'' = E\phi.$$

Write

$$k^2 = \frac{2mE}{\hbar^2}.$$

Then

$$\phi(x) = A\sin(kx) + B\cos(kx).$$

The boundary condition $\phi(0) = 0$ gives

$$B = 0.$$

The condition $\phi(L) = 0$ gives

$$\sin(kL) = 0.$$

Therefore

$$k = \frac{n\pi}{L}, \qquad n = 1, 2, \ldots.$$

Theorem 4.16.2 (Energy eigenbasis)**.**
The normalised eigenfunctions are

$$\boxed{\phi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right).}$$

Their energies are

$$\boxed{E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}.}$$

Moreover,

$$H\phi_n = E_n\phi_n.$$

Expand the initial state in the energy basis:

$$\psi(x,0) = \sum_{n=1}^{\infty} c_n(0)\phi_n(x),$$

where

$$c_n(0) = \langle \phi_n, \psi(\,\cdot\,,0)\rangle\,.$$

At time t, allow the coordinates to move:

$$\psi(x,t) = \sum_{n=1}^{\infty} c_n(t)\phi_n(x).$$

Substitute this into

$$\mathrm{i}\hbar\psi_t = H\psi.$$

The left-hand side is

$$\mathrm{i}\hbar \sum_n c_n'(t)\phi_n.$$

The right-hand side is

$$\sum_n E_n c_n(t)\phi_n.$$

Equality of coordinates gives

$$\mathrm{i}\hbar c_n'(t) = E_n c_n(t).$$

Theorem 4.16.3 (Time evolution in the well)**.**
Each energy coordinate evolves independently:

$$\boxed{c_n(t) = \mathrm{e}^{-\mathrm{i}E_n t/\hbar} c_n(0).}$$

Therefore

$$\boxed{\psi(x,t) = \sum_{n=1}^{\infty} c_n(0) \mathrm{e}^{-\mathrm{i}E_n t/\hbar} \phi_n(x).}$$

The Hamiltonian multiplies the nth coordinate by E_n.

Time evolution multiplies it by a phase:

$$\mathrm{e}^{-\mathrm{i}E_n t/\hbar}.$$

Its magnitude is one, so

$$|c_n(t)| = |c_n(0)|.$$

Worked example 4.16.4 (Two energy levels)**.**
Suppose

$$\psi(x,0) = \frac{1}{\sqrt{2}}\big(\phi_1(x) + \phi_2(x)\big).$$

Then

$$\psi(x,t) = \frac{1}{\sqrt{2}}\big(\mathrm{e}^{-\mathrm{i}E_1 t/\hbar}\phi_1(x) + \mathrm{e}^{-\mathrm{i}E_2 t/\hbar}\phi_2(x)\big).$$

The two coefficient magnitudes remain $1/\sqrt{2}$.

Only their phases change.

The relative phase is

$$\frac{(E_2 - E_1)t}{\hbar}.$$

4.17. Heat flow and quantum motion compared

Both equations become scalar equations in an eigenbasis.

	Heat equation	Schrödinger equation
Equation	$u_t = -Au$	$\mathrm{i}\hbar\psi_t = H\psi$
Coordinate	$c_n(t)$	$c_n(t)$
Evolution	$\mathrm{e}^{-\lambda_n t}c_n(0)$	$\mathrm{e}^{-\mathrm{i}E_n t/\hbar}c_n(0)$
Effect	Magnitudes decay	Magnitudes remain fixed
Meaning	Smoothing	Phase rotation

Basis direction	**Coordinate**	**Multiplier**
q_i	c_i	λ_i
$\mathrm{e}^{\mathrm{i}n\pi x/L}$	c_n	$\mathrm{i}n\pi/L$ or $(n\pi/L)^2$
$\mathrm{e}^{\mathrm{i}kx}$	$\widehat{f}(k)$	$\mathrm{i}k$ or k^2
$\sin(n\pi x/L)$	c_n	E_n

The repeated method is:

choose a basis adapted to the operator,
take coordinates,
multiply each coordinate,
reconstruct the vector.

Part V.

Synthesis

5. The Three-Part Dictionary

One idea, three index sets

The central message can be compressed into two lines:

$$\boxed{\text{finite indices} \longrightarrow \text{countable indices} \longrightarrow \text{continuous indices}},$$

$$\boxed{\sum_{i} \longrightarrow \sum_{n} \longrightarrow \sum_{x} \text{(heuristic)} \rightsquigarrow \int \mathrm{d}x}.$$

The table follows the common section numbering.

Table 5.1.: Parallel dictionary for finite vectors, sequences and functions

Concept	Part I: $\mathbb{R}^N$	Part II: ℓ^2	Part III: $L^2(I)$
Underlying vector	$v = (v_1, \ldots, v_N)$	$a = (a_1, a_2, \ldots)$	$f = (f_x)_{x \in I}$ initially; then $f(x)$
Index set	$i = 1, \ldots, N$	$n \in \mathbb{N}$	$x \in I \subseteq \mathbb{R}$
Natural space	$\mathbb{R}^N$	$\ell^2 = \{a : \sum_n a_n^2 < \infty\}$	$L^2(I) = \{f : \int_I \lvert f\rvert^2 < \infty\}$ modulo equality almost everywhere
Coordinate sum	$\sum_{i=1}^N$	$\sum_{n=1}^\infty$	$\sum_x$ only as motivation; rigorously $\int_I \mathrm{d}x$
Linear combination	$\sum_{j=1}^k \alpha_j v_j$	Finite algebraic combinations; limits add infinite expansions	Finite algebraic combinations; limits produce function expansions
Standard or useful basis	Standard vectors e_i	Standard sequences e_n	Problem-adapted basis functions, e.g. $\sqrt{2/\pi}\sin(nx)$ on $(0, \pi)$
Basis meaning	Finite algebraic basis	Complete orthonormal system with norm-convergent series	Complete orthonormal system with L^2-convergent series

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Table 5.1 continued

Concept	Part I: $\mathbb{R}^N$	Part II: ℓ^2	Part III: $L^2(I)$
Decomposition	$v = \sum_i v_i e_i$	$a = \sum_n a_n e_n$ in ℓ^2	$f = \sum_n c_n \phi_n$ in L^2, $c_n = \langle \phi_n, f \rangle$
New convergence issue	None: sums are finite	Series and truncation tails must converge	L^2 convergence need not imply pointwise convergence
Linear operator	$A : \mathbb{R}^N \to \mathbb{R}^N$	Bounded $A : \ell^2 \to \ell^2$, or unbounded A with a domain	Multiplication, differentiation and integral operators; domains and boundary conditions matter
Component action	$(Av)_i = \sum_j A_{ij} v_j$	$(Aa)_n = \sum_m A_{nm} a_m$	$(Af)_x = \sum_y (A_{xy} f_y) \rightsquigarrow \int_I A(x,y) f(y)\,\mathrm{d}y$
Matrix or kernel	Finite matrix (A_{ij})	Infinite matrix (A_{nm}) when its series defines an operator	Kernel $A(x,y)$, a continuously indexed matrix
Inner product	$\langle u, v \rangle = \sum_i u_i v_i$	$\langle a, b \rangle = \sum_n a_n b_n$	$\sum_x (f_x g_x) \rightsquigarrow \langle f, g \rangle = \int_I f(x) g(x)\,\mathrm{d}x$
p-norm	$\|v\|_p = (\sum_i \|v_i\|^p)^{1/p}$	$\|a\|_p = (\sum_n \|a_n\|^p)^{1/p}$	$\|f\|_p = (\int_I \|f\|^p)^{1/p}$

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Table 5.1 continued

Concept	Part I: $\mathbb{R}^N$	Part II: ℓ^2	Part III: $L^2(I)$
Norm equivalence	All p-norms equivalent on fixed $\mathbb{R}^N$; $C^{-1}\|v\|_p \le \|v\|_q \le C\|v\|_p$	Fails: block-vector norm ratios grow without bound; ℓ^p spaces differ	Fails: concentrating spikes have incompatible norm ratios; L^p spaces differ
Normalisation	$\widehat{v} = v/\|v\|_2$	$\widehat{a} = a/\|a\|_2$	$\widehat{f} = f/\|f\|_2$
Orthogonality	$\sum_i u_i v_i = 0$	$\sum_n a_n b_n = 0$	$\int_I f(x)g(x)\,\mathrm{d}x = 0$
Projection onto a unit basis element	$P_i v = \langle e_i, v\rangle\, e_i$	$P_n a = \langle e_n, a\rangle\, e_n$	$P_\phi f = \langle \phi, f\rangle\, \phi$
Projection kernel	$P_i = e_i e_i^T$	$(P_n)_{jk} = \delta_{jn}\delta_{kn}$	$P_\phi(x,y) = \phi(x)\phi(y)$
Resolution of identity	$I = \sum_{i=1}^N P_i$	$I = \sum_{n=1}^\infty P_n$ strongly	$I = \sum_n P_n$ in a complete basis; formally $I(x,y) = \delta(x-y)$
Parseval	$\|v\|_2^2 = \sum_i \|\langle e_i, v\rangle\|^2$	$\|a\|_2^2 = \sum_n \|\langle e_n, a\rangle\|^2$	$\|f\|_2^2 = \sum_n \|\langle \phi_n, f\rangle\|^2$
Eigen-equation	$Av = \lambda v$	$Aa = \lambda a$	$Af = \lambda f$
Eigen-object	Eigenvector	Eigenvector (sequence)	Eigenfunction

Continued on next page

Table 5.1 continued

Concept	Part I: $\mathbb{R}^N$	Part II: ℓ^2	Part III: $L^2(I)$
Eigenspace and degeneracy	$E_\lambda = \ker(A - \lambda I)$; finite dimension	Eigenspace may have finite or infinite dimension	Boundary conditions influence eigenspaces; symmetry can create degeneracy
Adjoint	$A^* = A^T$ in a real orthonormal basis	$\langle a, Ab\rangle = \langle A^*a, b\rangle$; domains matter if unbounded	$\langle f, Ag\rangle = \langle A^*f, g\rangle$; transpose the kernel or integrate by parts
Self-adjoint	Real symmetric matrix	$A = A^*$ with matching domains	$A = A^*$ with boundary conditions included in the domain
Eigenvalue consequence	Self-adjoint eigenvalues are real	Same	Same
Eigenvector consequence	Distinct eigenvalues give orthogonal eigenvectors	Same, when eigenvectors exist	Distinct eigenvalues give orthogonal eigenfunctions

Continued on next page

Table 5.1 continued

Concept	Part I: $\mathbb{R}^N$	Part II: ℓ^2	Part III: $L^2(I)$
Spectral theorem	Every self-adjoint operator has an orthonormal eigenbasis	Compact self-adjoint operators have a discrete eigenbasis; general operators may require spectral measures	Compact symmetric kernels often have eigenfunction bases; multiplication and differential operators may have continuous spectrum
Spectral decomposition	$A = \sum_\lambda \lambda P_\lambda$	$A = \sum_n \lambda_n P_n$ in the discrete case; generally $A = \int \lambda \, \mathrm{d}P(\lambda)$	Discrete eigenfunction sums or the spectral integral $A = \int \lambda \, \mathrm{d}P(\lambda)$
Characteristic new issue	Basis choice, but no analytical obstruction	Convergence, completeness, boundedness and domains	Integration, almost-everywhere equality, boundary conditions and continuous spectrum

A final worked bridge: one-dimensional projection

Choose a normalised direction in each setting:

$$e_i \in \mathbb{R}^N, \qquad e_n \in \ell^2, \qquad \phi \in L^2(I).$$

Projection is always "overlap times direction":

$$P_i v = \left(\sum_{j=1}^{N} (e_i)_j v_j \right) e_i,$$

$$P_n a = \left(\sum_{m=1}^{\infty} (e_n)_m a_m \right) e_n,$$

$$P_\phi f = \left(\int_I \phi(y) f(y)\, \mathrm{d}y \right) \phi.$$

In kernel notation the last line is

$$(P_\phi f)(x) = \int_I \underbrace{\phi(x)\phi(y)}_{P_\phi(x,y)} f(y)\, \mathrm{d}y.$$

Only the index set, and hence the summation rule, has changed.

> You have already seen the idea. Only the index set has changed.

Closing perspective

A function is a vector. A Fourier coefficient is a coordinate. An integral kernel is a matrix. A modal expansion is a spectral decomposition.

In infinite dimensions, ask:

- In which norm does a series or approximation converge?
- Is the space complete?
- Is the operator bounded, and if not, what is its domain?
- Which boundary conditions make the operator self-adjoint?
- Is the spectrum discrete, continuous, or a mixture of both?

These are the basic questions of functional analysis.

A. Deferred Proof

A.1. Equivalence of p-norms on $\mathbb{R}^N$

Proof of Theorem 1.11.1. It is enough to assume $1 \le p \le q \le \infty$.

First,

$$\|v\|_q \le \|v\|_p .$$

For finite q, Hölder's inequality gives

$$\sum_{i=1}^{N} |v_i|^p \le \left(\sum_{i=1}^{N} |v_i|^q \right)^{p/q} N^{1-p/q}.$$

Hence

$$\|v\|_p \le N^{1/p-1/q} \|v\|_q .$$

When $q = \infty$,

$$\|v\|_p^p \le N \|v\|_\infty^p .$$

Thus, for fixed N, both norms bound one another by constants independent of v. Interchanging p and q covers every pair. □